

The source boundary of extensional acyclic digraphs

An OEIS conjecture, finite-defect enumeration, and dyadic non-holonomicity

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In the mathematical research context of the ProveIt repository

Abstract

An extensional acyclic digraph has pairwise distinct out-neighborhoods. We give a self-contained proof of the formula $a(n) = n - \lceil \log_2 n \rceil$ for its maximum possible number of sources, the formula still marked conjectural in OEIS A182220 when consulted. The pigeonhole upper bound is already present in Tomescu's 2011 thesis; we credit that fact and do not present it as a new discovery.

The main development concerns the boundary of the source-count triangle A182162. If $u_m = A001192(m)$, $q = \lceil \log_2 n \rceil$, and $b_n^{(d)}$ counts isomorphism classes with exactly $n - q - d$ sources, we obtain an exact $(d + 1)$ -term formula and, in particular, $b_n^{(0)} = u_q \binom{2^q - q}{n - q}$. A uniform covering estimate then proves a dyadic entropy law for every fixed d . The complete set of subsequential exponential growth rates is an explicit interval, equal to $[1, 4]$ when $d = 0$. The exponential jumps across powers of two imply that none of these boundary sequences is P-recursive. Consequently their ordinary generating functions are not D-finite.

We also give an all-orders expansion of the Boolean-subset factor, retaining the small-core count exactly, and a total-variation binomial law for the number of missing arcs near a complete Boolean layer. The proofs are accompanied by independently implemented counting recurrences and exhaustive small-graph checks. These are conventional mathematical proofs with computational tests, not Lean-verified results. Historical priority for the boundary refinements is not established.

Primary sequences: A182220, A182162, A001192, A182161.

Keywords: transitive finite sets; source enumeration; Boolean layers; finite defects; entropy; polynomial recurrences; exact sampling.

[write] **Added 5 October 2026, batch 98: the two Parts.** Since 5 October 2026 this report has two Parts. Part I (Sections 1–13 and Appendices A and B) is the batch-85 manuscript described in the first note of Section 1, printed as before: no section, equation, theorem, table or label of it changed number. Part II (Sections 14–25) merges two independent batch-98 manuscripts that answer Part I’s first research question (Section 12.2): for every fixed defect d , the circle $|z| = R_d$ is a natural boundary of B_d (Theorem 14.2), with an explicit radial phase at every dyadic root of unity, a two-term radial growth law, an L^p comparison on every arc, natural boundaries for arithmetic sections, and further consequences. Both manuscripts are AI-assisted, unrefereed and unformalized; the provenance notes at the start of Part II give their sources, pins and the merge choices. Part II uses several letters (a , ρ , T , L , N , K , A_q , h) differently from Part I; its notation table fixes them. Dated notes at Remark 8.3, Question 1, the Conclusion and the proof-status ledger record the answer.

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Part I

The source boundary: finite-defect enumeration and dyadic non-holonomicity

1 Problem, contribution, and prior-art boundary

1.1 What is being answered

Let $a(n)$ be the largest possible number of sources in an extensional acyclic digraph on $n \geq 1$ vertices. OEIS A182220 identifies this quantity and records, among other equivalent conjectural expressions,

$$a(n) = n - 1 - \lfloor \log_2(n - 1) \rfloor \quad (n \geq 3). \quad (1)$$

The entry attributes this formula to Antti Karttunen, August 17, 2013 [1]. For $n \geq 2$, the right side equals $n - \lceil \log_2 n \rceil$; the latter also gives $a(1) = 1$. The problem is not an issue of numerical fitting. We prove both the universal obstruction and an explicit construction for every admissible source count.

The more substantial question is: *how many graphs lie on, or a fixed distance below, this extremal boundary, and how do those counts behave?* Write $u(n, k)$ for the number of isomorphism classes with n vertices and k sources. The labeled count is $n!u(n, k)$; the factor $n!$ will be justified, not assumed. Define

$$q = q(n) = \lceil \log_2 n \rceil, \quad b_n^{(d)} = u(n, n - q - d), \quad d \in \mathbb{N}_0, \quad (2)$$

with value zero if the requested number of sources is not positive. For each fixed d , this convention affects only finitely many terms.

1.2 What was already known

Peddicord studied the numbers u_m of finite transitive, or full, sets [5, 3]. Policriti and Tomescu developed source-sensitive enumeration of extensional acyclic digraphs [6]. Tomescu's thesis explicitly gives the source bound $k \leq n - \lceil \log_2 n \rceil$ on printed page 29, along with source-counting recurrences and rigidity results [7]. The source sieve used below is also encoded in Johnston's 2012 Maple program in A182162 [2]. Wagner studied asymptotic enumeration and distributions in the unrestricted graph class [8].

Accordingly, neither the upper bound, the correspondence with full sets, rigidity, nor the general source sieve is claimed as new. An OEIS conjectural label does not establish that a statement is an unsolved research problem in the literature. Our proof supplies a direct resolution of the displayed formula; its upper-bound half has explicit prior art.

The exact boundary classification, the finite-defect specialization, the uniform entropy profile, and the non-P-recursiveness theorem form the development of this article. Some exact identities are short consequences of the classical source sieve. We make no assertion of historical priority for those consequences or for the probabilistic refinements. The meaningful claim is that complete proofs are supplied here, not that an exhaustive literature search has certified novelty.

1.3 Repository relationship and reproducibility

The supplied ProveIt repository was inspected at commit

6bf7f30d0352f7596e70928b3d4f304914075907.

Its organization separates mathematical sources, research reports, and executable certificates [9]. Searches for the present OEIS identifier did not locate an existing treatment. That limited search is not an exhaustive duplication audit. No repository theorem is used as an unverified premise below, and no repository files were modified.

The accompanying `verify.py` uses exact integers and independently implements a marked-source sieve and a source-deletion recurrence. It also enumerates all extensional DAG isomorphism classes through seven vertices. The finite tests support error detection; the proofs, rather than the tests, establish the infinite statements.

[write] **Added 3 October 2026, batch 85.** *Provenance.* This report is manuscript 07 of batch 85 of ProveIt’s docs/incoming intake: the archive `OEIS_A182220_Source_Boundary_Research.zip` (wrapper directory `oeis_source_boundary/`), arrival commit `9d6968c8a`, placed in `ddf8df5d5` (batch 85C), written against the pin `6bf7f30d0` quoted above. It is a single-source report. Its title block names no author and no tool. The write phase prefixed every label with `sbd:`, added the dated notes marked [write], and kept the title page out of the PDF page anchors; no statement, proof or number was changed.

Priority. The formula $a(n) = n - \lfloor \log_2 n \rfloor$ is *not* presented as a new discovery. Its upper half is Tomescu’s: on printed page 29 of the thesis [7], which the intake read, the count of unlabeled EADs with n vertices and s sources is said to vanish unless $2^{n-s} \geq n$, equivalently $s \leq n - \lfloor \log_2 n \rfloor$, “by extensionality and the pigeonhole principle”. What the present proof adds is the matching construction for every admissible source count (the sufficiency half of Theorem 3.1), itself elementary; no priority is claimed for it. The word “conjecture” in OEIS records the status of an entry’s formula in the encyclopedia. It does not certify that the problem was open in the literature, nor that the proof below is the first.

OEIS check. On 3 October 2026 the intake read A182220 on <https://oeis.org/A182220>. Its formula section lists four conjectural expressions: Karttunen’s (1) (2013), and others by W. I. Hurt (2014), P. Barry (2017) and A. Krivilev (August 2026, the “formula from August 2026” of the package’s `SOURCES.md`). Each is elementarily equivalent to (5) (Barry’s counts the powers of two in $[1, n - 1]$, Krivilev’s the non-powers of two there), and the intake confirmed all four against (5) for $n < 600$, as well as the 30 displayed terms. Theorem 3.1 therefore settles all four.

Repository. The claims of this subsection still hold at the time of writing: no other file of the repository mentions A182220, A182162, A001192, A182161, extensional acyclic digraphs, Tomescu, Policriti or Peddicord. No Lean or Rocq declaration formalizes any statement of this report, and its place in the research-report collection confers no formal status; Section 12.1 is a plan. Related material, none of it a shared theorem: (i) Lemma 2.1 is the classical Mostowski collapse, which the surreal report `Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets` proves for rooted well-founded codes as its `hset:lem:collapse`; (ii) the collection report `generating-functions-and-asymptotics/oeis-sequence-asymptotics/a116379-bounded-identity-trees` counts rooted identity trees, the other classical coding of hereditarily finite sets (it cites Meir, Moon and Mycielski); and (iii) Lemma 8.1 is a counterpart of a criterion of `a003407-dyadic-scaling-rigidity`, discussed in a note after it.

Notation. Several letters carry more than one local meaning:

- d is the defect, never the differential in d/dp or d/dz ; $B_d(z)$ in (33) is a generating function, while B_{2j} in Theorem 9.1 are Bernoulli numbers.
- $T = 2^m$ in Sections 7 and 9, but $T(n, k)$ in Appendix A is the labeled count $n! u(n, k)$ of A182162, as is the symbol $E(n, a(n))$ in (11), which the text does not define otherwise.
- C is a transitive core (and C, A constants in the proof of Lemma 8.1; $C_1(m, p)$ a correction coefficient); L is a source family, $L(z)$ the lacunary series of Proposition 3.2, and L also a truncation order in Theorem 9.1.
- $N = 2^m$ in the fixed-deletion and arc sections, N_r the jump indices of Lemma 8.1, $N(0, 1)$ the normal law; $p = n/T$ is a phase, p_j the recurrence polynomials of (32); R a removed family, R_d a radius; $M = 2^m - m$ and, in Theorem 6.1, $A = u_m \binom{M}{k}$.
- $\rho_n \in (1/2, 1]$ is the dyadic phase $n/2^q$, whereas $p = n/T$ with $T = 2^{q+d}$ lies in $(2^{-d-1}, 2^{-d}]$: the tempting reading “ $p = \rho_n$ ” is right only for $d = 0$.

No symbol was renamed.

2 Definitions and finite-set foundations

2.1 Orientation conventions

A digraph in this article is finite, loopless, and has no parallel arcs. An arc $v \rightarrow w$ means that w is an out-neighbor of v . A source has indegree zero; a sink has outdegree zero. A digraph is *extensional* when the map

$$v \mapsto N^+(v) = \{w : v \rightarrow w\}$$

is injective. It is acyclic when it contains no directed cycle. We abbreviate “extensional acyclic digraph” to EAD. Every nonempty EAD has exactly one sink: a DAG has a sink, and two sinks would have the same empty out-neighborhood. No transitive closure or transitive reduction of the arc relation is implicit in this definition.

Let u_m be the number of EAD isomorphism classes with m vertices, including $u_0 = 1$. We shall identify it with A001192. We use $\binom{M}{r} = 0$ if $r < 0$ or $r > M$ for nonnegative integer M . All logarithms without a subscript are natural.

2.2 Collapse, transitivity, and rigidity

Lemma 2.1 (Finite collapse). *For every EAD D , the recursion*

$$\phi(v) = \{\phi(w) : w \in N^+(v)\}$$

defines an injective map into the hereditarily finite sets. Its image $F = \{\phi(v) : v \in V(D)\}$ is a finite transitive set, and D is isomorphic to the membership digraph on F , with arcs $x \rightarrow y$ iff $y \in x$. Conversely, every finite transitive set yields an EAD this way. This gives a bijection between isomorphism classes of EADs and finite transitive sets.

Proof. Order the vertices so that every out-neighbor precedes its parent. The recursion is then finite and well defined. To prove injectivity, induct on the maximum height of two vertices, where height is the maximum length of a directed path from the vertex. Equal collapsed sets at height at most h have the same collapsed members. By the induction hypothesis the corresponding vertices of smaller height are identical. Thus their out-neighborhoods coincide, and extensionality identifies the two vertices. The base case is the unique sink.

Every member of $\phi(v)$ is another element of the image, so F is transitive: $\bigcup F \subseteq F$. Injectivity shows that membership reproduces exactly the arcs. Conversely, on a finite transitive set, Foundation precludes membership cycles, and Extensionality says that distinct elements have distinct member sets. These two constructions are inverse, including at the empty set. $\square$

Corollary 2.2 (Rigidity). *Every EAD has trivial automorphism group. Consequently the number of labeled EADs with n vertices and k sources is $n!u(n, k)$, and the total number is $n!u_n$.*

Proof. An automorphism preserves the recursive collapsed value of every vertex. The collapse is injective, so the automorphism fixes every vertex. There are therefore exactly $n!$ different labelings of each isomorphism class, and source count is preserved by relabeling. $\square$

This explains the normalization relating A182162 to $u(n, k)$ and A182161 to A001192 [2, 4, 3].

Lemma 2.3 (A sufficient bound on the small-core count). *For every $m \geq 0$,*

$$1 \leq u_m \leq 2^{\binom{m}{2}}. \quad (3)$$

Proof. The finite ordinal $m = \{0, 1, \dots, m-1\}$ is a transitive set of size m , so at least one class exists. Every DAG has a vertex ordering in which all arcs point to earlier vertices. For a fixed such ordering there are $2^{\binom{m}{2}}$ possible arc sets. Their isomorphism classes cover all DAG isomorphism classes, and hence all EAD classes. Injectivity of this covering is not needed for the upper bound. $\square$

3 The exact source maximum and all feasible source counts

Theorem 3.1 (Complete source support). *For $n \geq 1$ and $k \geq 1$, an EAD on n vertices with exactly k sources exists if and only if*

$$k \leq n, \quad n \leq 2^{n-k}. \quad (4)$$

In particular,

$$\boxed{a(n) = n - \lceil \log_2 n \rceil}. \quad (5)$$

Every integer from 1 through $a(n)$ occurs, so A182162 has no holes in the positive support of any row.

Proof. Suppose D has k sources and let C be its set of nonsources, $m = |C| = n - k$. No arc enters a source. Thus every out-neighborhood is a subset of C . The injective map $v \mapsto N^+(v)$ takes n vertices into a set of size 2^m , proving $n \leq 2^m$.

For sufficiency, first take $m = 0$. The conditions force $n = k = 1$, and the one-vertex edgeless digraph works. Suppose $m \geq 1$. Take a directed path on m core vertices, with arcs from each vertex to the preceding one. Its m out-neighborhoods are distinct. The full set C is not among them, since it would contain the vertex whose neighborhood it was, creating a loop. There are $2^m - m$ unused subsets of C , at least k by the assumed inequality. Choose k distinct unused subsets, one of them equal to C , and add a new vertex for each chosen subset, with exactly that subset as its out-neighborhood.

There are no arcs into the new vertices, so they are sources. All core vertices receive an arc from the new vertex whose neighborhood is C , so none is a source. Acyclicity and extensionality are immediate from the construction. Hence the graph has exactly k sources.

Finally, $n \leq 2^{n-k}$ is equivalent to $k \leq n - q(n)$. For $n \geq 1$, this upper endpoint is at least one. This proves both the maximum and the entire support assertion. $\square$

3.1 Elementary consequences for A182220

For $n \geq 2$, $\lceil \log_2 n \rceil = 1 + \lfloor \log_2(n-1) \rfloor$, which proves (1). Also,

$$a(n+1) - a(n) = 1 - \mathbf{1}_{\{n \text{ is a power of } 2\}}, \quad a(1) = 1. \quad (6)$$

This accounts for the repeated values. In particular the initial value 1 occurs three times, because both 1 and 2 are powers of two. For an interpretation by counting nonpowers of two, zero must be included:

$$a(n) = \#\{0 \leq j \leq n-1 : j \text{ is not a positive power of } 2\},$$

where “a power of 2” includes $2^0 = 1$. Omitting zero would create an off-by-one error. We do not assert every ancillary OEIS cross-reference without checking its conventions; the explicit formulas above remove that ambiguity.

3.2 A lacunary generating function for the maximum

The next consequence is included to separate the simple maximum sequence from the more intricate sequence counting its extremizers.

Proposition 3.2. *For $|z| < 1$,*

$$\sum_{n \geq 1} a(n)z^n = \frac{z}{(1-z)^2} - \frac{z}{1-z} \sum_{j \geq 0} z^{2^j}. \quad (7)$$

The unit circle is a natural boundary for this generating function.

Proof. Multiply the left side by $1-z$ and apply (6) to obtain the identity. Let $L(z) = \sum_{j \geq 0} z^{2^j}$ and let ζ be a root of unity of order 2^s . For all $j \geq s$, $\zeta^{2^j} = 1$; hence

$$L(r\zeta) = O(1) + \sum_{j \geq s} r^{2^j}, \quad r \uparrow 1,$$

and its real part tends to infinity. At every such $\zeta \neq 1$, the rational factor $z/(1-z)$ tends to a nonzero finite value, so the right side of (7) cannot extend holomorphically through ζ . These roots are dense on the circle. Analytic continuation across any point would give a neighborhood containing one of them, a contradiction. $\square$

The proof uses only a dense set of explicit singularities; no general lacunary-series theorem is needed.

4 Boolean-core classification and exact extremizer counts

4.1 The core is intrinsic

Let F be the collapsed transitive set of a graph. Its nonsources are exactly the elements which occur as members of other elements of F . Thus the nonsource core is

$$C = \bigcup F. \quad (8)$$

It is itself transitive. Indeed, $x \in y \in C$ means that $y \in z$ for some $z \in F$; then $y \in F$, and therefore $x \in \bigcup F = C$. If $L = F \setminus C$ is the set of sources, then

$$F = C \cup L, \quad L \subseteq \mathcal{P}(C) \setminus C. \quad (9)$$

The last exclusion uses transitivity of C , so that $C \subseteq \mathcal{P}(C)$. Conversely, (9) always defines a transitive set, but its nonsource core equals C only if $\bigcup(C \cup L) = C$. That covering condition is the sole extra constraint.

Lemma 4.1 (Automatic covering above half density). *Let C be any transitive set of size $m \geq 1$. If*

$$L \subseteq \mathcal{P}(C) \setminus C, \quad |L| = k, \quad m + k > 2^{m-1},$$

then $\bigcup(C \cup L) = C$.

Proof. If some $c \in C$ were outside the union, every member of the family $C \cup L$ would avoid c . All $m+k$ distinct subsets would then lie in $\mathcal{P}(C \setminus \{c\})$, which contains only 2^{m-1} subsets. $\square$

Theorem 4.2 (Complete classification of extremizers). *Let $n \geq 1$, $q = \lceil \log_2 n \rceil$, and $k = n - q$. The EAD isomorphism classes attaining $a(n)$ sources are in bijection with pairs*

$$C \in \mathcal{U}_q, \quad L \in \binom{\mathcal{P}(C) \setminus C}{k}, \quad (10)$$

where $\mathcal{U}_q$ is the set of transitive sets of size q . Consequently

$$\boxed{b_n^{(0)} = u_q \binom{2^q - q}{n - q}, \quad E(n, a(n)) = n! u_q \binom{2^q - q}{n - q}}. \quad (11)$$

Proof. Every extremizer has q nonsources by Theorem 3.1, and (8)–(9) recover its unique pair (C, L) . Conversely, if $q \geq 1$, then $n > 2^{q-1}$, so Lemma 4.1 shows that every pair in (10) has precisely C as its core and precisely L as its source set. No two different pairs give the same transitive set, since the core is its union. Each core offers exactly $2^q - q$ available subsets. Rigidity supplies the labeled factor. For $n = 1$, $q = 0$, and the single pair is $C = \emptyset$, $L = \{\emptyset\}$, giving the same formula. $\square$

Corollary 4.3 (Complete layers and sparse deletions). *For $m \geq 1$ and $0 \leq r < 2^{m-1}$,*

$$b_{2^m - r}^{(0)} = u_m \binom{2^m - m}{r}. \quad (12)$$

In particular $b_{2^m}^{(0)} = u_m$, and every extremizer on 2^m vertices is the membership graph of $\mathcal{P}(C)$ for a unique transitive m -element set C .

Proof. Complement the chosen source family in $\mathcal{P}(C) \setminus C$. The omitted family has size r , and at $r = 0$ no subsets are omitted. $\square$

Example 4.4. For $n = 8$ we have $q = 3$, $u_3 = 2$, and $b_8^{(0)} = 2$. For $n = 9$ the core size becomes four, and

$$b_9^{(0)} = u_4 \binom{12}{5} = 9 \cdot 792 = 7128.$$

The jump is not a numerical accident: the minimum allowed core size has increased, making a much larger Boolean layer available. Later sections quantify this phenomenon at every power of two.

5 The marked-source sieve and finite-defect diagonals

5.1 A classical identity, proved in the present normalization

Proposition 5.1 (Marked sources). *For $0 \leq t \leq n$,*

$$\sum_{s=t}^n \binom{s}{t} u(n, s) = u_{n-t} \binom{2^{n-t} - (n-t)}{t}. \quad (13)$$

For $n > 0$ we set $u(n, 0) = 0$; for $n = 0$ we set $u(0, 0) = 1$.

Proof. The left side counts an EAD isomorphism class together with a chosen t -element subset of its sources. Rigidity ensures that different chosen subsets cannot be identified by a nontrivial

automorphism. Delete the marked sources. Out-neighborhoods of the remaining vertices are unchanged, since no arc entered a source, so the remaining graph is an EAD on $n - t$ vertices.

Conversely, choose any such graph and choose t distinct subsets of its vertex set that are not already its out-neighborhoods. There are $2^{n-t} - (n - t)$ available subsets. Adjoin one marked source for each chosen subset. Existing sources are allowed to remain sources, which is why no covering constraint occurs in this construction. Deleting the marked sources reverses it. This proves the identity. $\square$

The identity is the marked-source version of the classical enumeration underlying the OEIS source sieve [2]. We use it as a proved starting point rather than a new counting principle.

[write] **Added 7 October 2026, batch 108 (reciprocal note).** The same sieve, with strongly connected components in place of source vertices, proves Robinson's component transform in the collection's report `a089479-fixed-permanent-matrices` (its Section 4, equation (19), label `fpm: eq:robinson`): a subset of the source components is marked with weight -1 , and every nonempty condensation has a source, so the signed sum vanishes. There it counts labelled digraphs whose strong components carry a multiplicative weight (the permanent of I plus the adjacency matrix). At permanent one these are exactly the labelled acyclic digraphs (OEIS A003024), and that report's permanent-one case is their classical asymptotic $A003024(n) \sim n! 2^{\binom{n}{2}} / (a\rho^n)$ with $\rho = 1.48807\dots$ (A245654) and $a = 0.57436\dots$ (A245655). The extensional acyclic digraphs of this report, counted here up to isomorphism (labelled count $n! u(n, k)$ by Corollary 2.2), form a subclass. That report imposes no extensionality condition, and nothing in this report is changed by it.

Corollary 5.2 (Binomial inversion). *For $1 \leq k \leq n$,*

$$u(n, k) = \sum_{t=k}^n (-1)^{t-k} \binom{t}{k} u_{n-t} \binom{2^{n-t} - (n-t)}{t}. \quad (14)$$

Also

$$u_0 = 1, \quad u_n = \sum_{m=0}^{n-1} (-1)^{n-m-1} \binom{2^m - m}{n-m} u_m \quad (n \geq 1). \quad (15)$$

Proof. If $P_n(z) = \sum_s u(n, s) z^s$, then (13) is equivalent to

$$P_n(z) = \sum_{t=0}^n u_{n-t} \binom{2^{n-t} - (n-t)}{t} (z-1)^t.$$

Extract z^k to get (14). For $n > 0$, evaluate at $z = 0$, use $P_n(0) = 0$, and isolate the term $t = 0$ to get (15). $\square$

5.2 Only finitely many correction terms survive

Theorem 5.3 (Exact finite-defect formula). *Fix $n \geq 1$ and $d \geq 0$ with $k = n - q - d \geq 1$, and put $m = q + d$. Then*

$$b_n^{(d)} = \sum_{j=0}^d (-1)^j \binom{k+j}{j} u_{m-j} \binom{2^{m-j} - (m-j)}{k+j}. \quad (16)$$

Thus an entry at distance d below the source boundary requires exactly $d + 1$ displayed terms, using total counts only at the logarithmic indices $q, q + 1, \dots, q + d$.

Proof. Substitute $t = k + j$ in (14). A potentially nonzero term must satisfy

$$k + j \leq 2^{m-j} - (m - j), \quad \text{equivalently} \quad n \leq 2^{m-j}.$$

Since $q = \lceil \log_2 n \rceil$, this requires $m - j \geq q$, or $j \leq d$. The surviving terms are exactly those displayed. All have a positive binomial coefficient in the stated range, though signs alternate. $\square$

For $d = 0$, (16) recovers the exact extremizer count, so that count is also an immediate boundary specialization of the known sieve. At distance one it becomes

$$b_n^{(1)} = u_{q+1} \binom{2^{q+1} - q - 1}{n - q - 1} - (n - q) b_n^{(0)}. \quad (17)$$

At distance two, writing $k = n - q - 2$,

$$\begin{aligned} b_n^{(2)} &= u_{q+2} \binom{2^{q+2} - q - 2}{k} - (k + 1) u_{q+1} \binom{2^{q+1} - q - 1}{k + 1} \\ &\quad + \binom{k + 2}{2} u_q \binom{2^q - q}{k + 2}. \end{aligned} \quad (18)$$

These identities should not be mistaken for asymptotic expansions: they are exact integer formulas, even when early terms nearly cancel.

6 Uniform covering estimates and random cores

6.1 An estimate valid away from the extremal boundary as well

Theorem 6.1 (Uniform core-covering bound). *Let $m \geq 0$, $k \geq 1$, $n = m + k$, and $k \leq M = 2^m - m$. Set $A = u_m \binom{M}{k}$. Then*

$$0 \leq A - u(n, k) \leq m 2^{-k} A. \quad (19)$$

For $m \geq 1$, a sharper upper bound on the relative deficit is

$$\frac{A - u(n, k)}{A} \leq \frac{\binom{2^{m-1} - m}{k}}{\binom{2^m - m}{k}} \frac{\sum_s s u(m, s)}{u_m}. \quad (20)$$

Proof. For a fixed transitive core C of size m , choose a uniform k -subset L of $\mathcal{P}(C) \setminus C$. The resulting graph is counted by $u(n, k)$ exactly when $\bigcup(C \cup L) = C$. Every element already in $\bigcup C$ is covered. If $c \in C \setminus \bigcup C$, then none of the m members of C contains c . Of the 2^{m-1} subsets avoiding c , precisely m are therefore forbidden core subsets. The probability that no chosen subset contains c is

$$\frac{\binom{2^{m-1} - m}{k}}{\binom{2^m - m}{k}} \leq 2^{-k}.$$

If the numerator vanishes the assertion is immediate. Otherwise compare successive factors of the two falling factorials; each ratio is at most $1/2$. Apply the union bound over the sources of C , whose number is at most m , and sum over all u_m cores. The sharper average-source version follows before replacing the number of sources by m . For $m = 0$, the only admissible choice is $k = 1$, and equality holds. $\square$

The average core-source count in (20) is itself exact: Proposition 5.1, with $t = 1$, gives

$$\frac{\sum_s s u(m, s)}{u_m} = \frac{(2^{m-1} - m + 1)u_{m-1}}{u_m} \quad (m \geq 1). \quad (21)$$

In applications where $k - \log_2 m \rightarrow +\infty$, the simpler bound already proves

$$u(m + k, k) = u_m \binom{2^m - m}{k} (1 + O(m2^{-k})). \quad (22)$$

The implicit error here is one-sided: the true value never exceeds the main expression.

6.2 Uniform sampling and the core law

There is an exact rejection sampler for a uniformly random isomorphism class with $n = m + k$ vertices and k sources. First draw a uniform core from $\mathcal{U}_m$, then a uniform k -subset of its M available neighborhoods. Keep the pair if it covers the core; otherwise repeat. Before rejection, every pair has exactly the same probability, and every accepted graph corresponds to one pair. Thus the accepted graph is exactly uniform. If $m2^{-k} < 1$, the expected number of trials is at most $(1 - m2^{-k})^{-1}$. This is conditional on having a procedure for uniformly sampling the much smaller core; it is not a claim of a complete polynomial-time sampler for arbitrary m .

At the extremal boundary, rejection is never necessary, and the core of a uniform extremizer is *exactly* uniform on $\mathcal{U}_q$. More generally, write δ for the actual rejection probability. Conditioning a finite probability distribution on an event of probability $1 - \delta$ changes it in total variation by exactly δ . Taking the core marginal cannot increase total variation. Therefore

$$d_{\text{TV}}(\text{core law of a uniform } (n, k) \text{ EAD}, \text{Unif}(\mathcal{U}_m)) \leq m2^{-k}. \quad (23)$$

This bound is uniform over all core structures. It explains why small-core statistics can be transferred to the high-source part of the triangle.

7 The dyadic entropy law

7.1 Uniform logarithmic asymptotics

Define the binary entropy, in natural units, by

$$H(p) = -p \log p - (1 - p) \log(1 - p), \quad H(0) = H(1) = 0.$$

For a fixed defect d , set

$$\rho_n = \frac{n}{2^{q(n)}} \in (1/2, 1], \quad \Phi_d(\rho) = \frac{2^d}{\rho} H\left(\frac{\rho}{2^d}\right). \quad (24)$$

The strict lower endpoint for ρ_n does not preclude subsequences converging to $1/2$.

Theorem 7.1 (Entropy profile). *For every fixed $d \geq 0$, as $n \rightarrow \infty$,*

$$\boxed{\log b_n^{(d)} = n\Phi_d(\rho_n) + O_d((\log n)^2)}. \quad (25)$$

The bound is uniform over the entire dyadic block, including its endpoints. No asymptotic formula for u_m is needed.

Proof. Let $m = q + d$, $T = 2^m$, and $k = n - m$. These parameters satisfy $m = O_d(\log n)$, $n \leq T < 2^{d+1}n$, and $k = n - O_d(\log n)$. Theorem 6.1 gives

$$\log b_n^{(d)} = \log u_m + \log \binom{T-m}{n-m} + O_d(m2^{-k}). \quad (26)$$

The small-core logarithm is $O_d((\log n)^2)$ by Lemma 2.3. The exact identity

$$\binom{T-m}{n-m} = \binom{T}{n} \frac{(n)_m}{(T)_m} \quad (27)$$

shows that the two binomial logarithms differ by $O_d(m)$: every factor $(n-i)/(T-i)$ is bounded below by a positive constant depending only on d , for all sufficiently large n .

For completeness, the elementary entropy bounds are

$$\frac{e^{TH(n/T)}}{T+1} \leq \binom{T}{n} \leq e^{TH(n/T)}. \quad (28)$$

For $0 < n < T$, consider a $\text{Bin}(T, n/T)$ random variable. Its probability at n is a maximum probability, hence is at least $1/(T+1)$ and at most one. Rearranging gives (28); the endpoint cases are immediate. It follows that $\log \binom{T}{n} = TH(n/T) + O(\log T)$. Finally $TH(n/T) = n\Phi_d(\rho_n)$. Combining these estimates proves (25), with no loss of uniformity as $n/T \rightarrow 1$. $\square$

7.2 Every subsequential exponential rate

Set

$$\lambda_d = 2^d H(2^{-d}), \quad d \geq 0. \quad (29)$$

In particular $\lambda_0 = 0$, $\lambda_1 = \log 4$, and $e^{\lambda_2} = 256/27$.

Theorem 7.2 (Complete growth-rate interval). *For each fixed $d \geq 0$, the set of subsequential limits of $(b_n^{(d)})^{1/n}$ is exactly*

$$[e^{\lambda_d}, e^{\lambda_{d+1}}]. \quad (30)$$

Thus

$$\liminf_{n \rightarrow \infty} (b_n^{(0)})^{1/n} = 1, \quad \limsup_{n \rightarrow \infty} (b_n^{(0)})^{1/n} = 4. \quad (31)$$

No fixed-defect boundary sequence has a single exponential growth constant.

Proof. Write $p = \rho/2^d$. Then $\Phi_d(\rho) = H(p)/p$. For $0 < p < 1$,

$$\frac{d}{dp} \frac{H(p)}{p} = \frac{pH'(p) - H(p)}{p^2} = \frac{\log(1-p)}{p^2} < 0.$$

Hence Φ_d is strictly decreasing, with values λ_{d+1} at $\rho = 1/2$ and λ_d at $\rho = 1$. Every $\rho \in (1/2, 1)$ is the limit of phases obtained by rounding $\rho 2^q$ to integers. Powers $n = 2^q$ realize phase one, while $n = 2^{q-1} + 1$ tends to phase $1/2$. The entropy theorem gives the entire claimed interval. Conversely, compactness of $[1/2, 1]$ and continuity of Φ_d exclude any other subsequential limits. $\square$

In particular the first three intervals, for $d = 0, 1, 2$, are

$$[1, 4], \quad [4, 256/27], \quad [256/27, 2^{24}/7^7].$$

The upper endpoint for one defect equals the lower endpoint for the next. This expresses the fact that crossing a dyadic boundary adds one core vertex, just as increasing the allowed source deficit does.

8 Non-P-recursiveness of every fixed-defect boundary

A sequence (c_n) is P-recursive if, for all sufficiently large n ,

$$\sum_{j=0}^h p_j(n) c_{n+j} = 0 \quad (32)$$

for polynomials p_j , not all zero. Allowing a finite exceptional initial segment makes the conclusion below stronger, not weaker. A formal power series is D-finite if it satisfies a nonzero homogeneous linear differential equation with polynomial coefficients.

Lemma 8.1 (An exponential-jump obstruction). *Suppose a nonnegative sequence (c_n) has integers $N_r \rightarrow \infty$ and constants $\alpha < \beta$ such that, for every fixed $j \geq 0$,*

$$c_{N_r-j} = \exp(\alpha N_r + o(N_r)), \quad c_{N_r+1} = \exp(\beta N_r + o(N_r)).$$

Then (c_n) is not P-recursive.

Proof. Suppose (32) holds, and choose h as the largest index with $p_h \neq 0$. If $h = 0$, positivity along (N_r) contradicts the recurrence. Otherwise apply it at $n = N_r + 1 - h$. For sufficiently large r , $p_h(n)$ is nonzero, and ratios of fixed polynomials give a bound

$$c_{N_r+1} \leq C N_r^A \sum_{j=0}^{h-1} c_{N_r+1-h+j} = \exp(\alpha N_r + o(N_r))$$

for constants C, A . This contradicts $\beta > \alpha$. Absolute values of the coefficients were used, so cancellation cannot invalidate the estimate. The argument also applies to complex polynomial coefficients. $\square$

[write] **Added 3 October 2026, batch 85.** The collection report `generating-functions-and-asymptotics/oeis-sequence-asymptotics/a003407-dyadic-scaling-rigidity` proves, in its Part II, an obstruction that runs the other way. Its Corollary `dsr:nh:cor:jumps` (a consequence of its local-growth rigidity theorem `dsr:nh:thm:rigidity`, which rests on a G-function input) says that an eventually positive, exponentially bounded, integer P-recursive sequence without an n th-root limit must have $u_{n+1}/u_n > e^{\varepsilon n}$ for infinitely many n : upward exponential jumps are *necessary* for such oscillation, and their absence rules P-recursiveness out. Lemma 8.1 is a *sufficient* condition in the opposite situation: a jump preceded by a plateau of every fixed length at the lower rate rules P-recursiveness out. It needs neither integrality nor an exponential bound, and its proof is elementary. The two criteria are complementary, not competing. The boundary sequences $b^{(d)}$ do have the upward jumps (from $e^{\lambda_d N_r}$ to $e^{\lambda_{d+1} N_r}$ at $N_r = 2^r$), so the a003407 criterion cannot exclude their P-recursiveness, and Lemma 8.1 is the one that applies. Conversely, the 3AP-free permutation counts of a003407 have only subexponential forward increases, so Lemma 8.1 does not apply to them. The plateau hypothesis “for every fixed $j \geq 0$ ” is essential: a003407’s example $3^n + (-3)^n + 2^n$, which satisfies a constant-coefficient recurrence, jumps from rate $\log 2$ at odd indices to $\log 3$ at even ones, but its two-term window never stays at the lower rate.

Theorem 8.2 (Non-holonomicity). *For every fixed $d \geq 0$, neither $(b_n^{(d)})$ nor $(n!b_n^{(d)})$ is P-recursive. The ordinary generating function*

$$B_d(z) = \sum_{n \geq 1} b_n^{(d)} z^n \quad (33)$$

is not D-finite. Its radius of convergence is

$$R_d = e^{-\lambda_{d+1}}; \quad R_0 = \frac{1}{4}. \quad (34)$$

Proof. Take $N_r = 2^r$. For every fixed $j \geq 0$, the index $N_r - j$ remains in the dyadic block with $q = r$ when r is large, and its phase tends to one. By Theorem 7.1,

$$\log b_{N_r-j}^{(d)} = \lambda_d N_r + o(N_r).$$

At $N_r + 1$, $q = r + 1$ and the phase tends to $1/2$, giving

$$\log b_{N_r+1}^{(d)} = \lambda_{d+1} N_r + o(N_r).$$

The strict decrease proved in Theorem 7.2 yields $\lambda_{d+1} > \lambda_d$. Lemma 8.1 applies.

If $n!b_n^{(d)}$ satisfied a polynomial recurrence, division by $n!$ would give one for $b_n^{(d)}$: each factor $(n+j)!/n!$ is a polynomial in n . Thus the labeled sequence is also non-P-recursive.

For completeness, extracting coefficients from a nonzero polynomial linear differential equation for B_d yields a nonzero finite-shift polynomial recurrence for its coefficients. Indeed, coefficients of $z^i(d/dz)^j B_d$ are shifted coefficients multiplied by falling factorials in the index; a common shift removes any negative indices. This contradicts what was just proved. Finally, the Cauchy–Hadamard formula and Theorem 7.2 give (34). $\square$

Remark 8.3 (What the theorem does not say). Non-D-finiteness does not by itself imply that the circle $|z| = R_d$ is a natural boundary, nor does it rule out useful nonlinear functional equations. We prove a natural boundary for the maximum sequence in Proposition 3.2, but do not make that stronger claim for B_d . The labeled ordinary generating series has radius zero; the series B_d is also the exponential generating series of the labeled boundary counts.

[write] **Added 5 October 2026, batch 98.** The stronger claim is now proved, in Part II: for every fixed $d \geq 0$ the circle $|z| = R_d$ is a natural boundary of B_d (Theorem 14.2, proved independently by both Part II manuscripts), and B_d is unbounded, faster than every power of $1/t$, on the radius to every point $R_d\zeta$ with ζ a root of unity of power-of-two order. The natural boundary gives a second route to the non-D-finiteness of Theorem 8.2 and adds non-algebraicity (Corollary 20.1). The remark’s other caveat stands: nonlinear functional or differential equations are not excluded, and Part II lists this as an open question (Section 24).

9 An all-orders expansion inside a dyadic block

The entropy law is uniform even at complete layers. More detailed prefactors require staying away from the degeneracy $p = 1$. This section retains u_m exactly; it does not purport to replace the classical, separate asymptotic analysis of the total EAD count.

Theorem 9.1 (Explicit expansion with exact core amplitude). *Fix $d \geq 0$ and $\varepsilon > 0$. Let*

$$m = q + d, \quad T = 2^m, \quad p = n/T, \quad k = n - m, \quad \varepsilon \leq p \leq 1 - \varepsilon.$$

Define $S_\ell(m) = \sum_{i=0}^{m-1} i^\ell$ and let B_{2j} denote the Bernoulli numbers with $B_2 = 1/6$. For fixed integers $L \geq 0$, $J \geq 1$,

$$\begin{aligned} \log b_n^{(d)} &= \log u_m + TH(p) - \frac{1}{2} \log(2\pi T p(1-p)) + m \log p \\ &\quad - \sum_{\ell=1}^L \frac{p^{-\ell} - 1}{\ell T^\ell} S_\ell(m) \\ &\quad + \sum_{j=1}^J \frac{B_{2j}}{2j(2j-1)T^{2j-1}} [1 - p^{1-2j} - (1-p)^{1-2j}] \\ &\quad + O_{d,\varepsilon,L,J} \left(\frac{m^{L+2}}{T^{L+1}} + T^{-(2J+1)} + m2^{-k} \right). \end{aligned} \tag{35}$$

The first sum is empty when $L = 0$.

Proof. Start with (26) and (27). The logarithm of the falling-factorial ratio is

$$m \log p + \sum_{i=0}^{m-1} \left[\log \left(1 - \frac{i}{pT} \right) - \log \left(1 - \frac{i}{T} \right) \right].$$

Since $m = O_d(\log T)$ and $p \geq \varepsilon$, expand both logarithms through degree L . The coefficient of $T^{-\ell}$ is exactly the first sum in (35). Summing the elementary Taylor remainders gives $O(m^{L+2}/T^{L+1})$.

Apply the Stirling expansion for $\log(T!)$, $\log((pT)!)$, and $\log(((1-p)T)!)$. Subtraction produces $TH(p)$, the displayed Gaussian prefactor, and the Bernoulli sum. Its remainder after J terms is $O_{\varepsilon,J}(T^{-(2J+1)})$. This standard expansion follows, for example, by Euler–Maclaurin summation of $\log x$; its remainder has the asserted bound because pT and $(1-p)T$ are both at least εT . Finally include the covering error from (26). $\square$

In particular,

$$b_n^{(d)} = \frac{u_m p^m e^{TH(p)}}{\sqrt{2\pi T p(1-p)}} \left[1 + \frac{C_1(m, p)}{T} + O_{d,\varepsilon} \left(\frac{m^4}{T^2} \right) \right], \quad (36)$$

$$C_1(m, p) = -\frac{1-p}{2p} m(m-1) + \frac{1-p^{-1} - (1-p)^{-1}}{12}.$$

Taking larger L and J supplies arbitrarily many logarithmic corrections. Exponentiating gives a multiplicative expansion whose coefficients are explicit polynomials in m , with rational functions of p as coefficients.

For $d \geq 1$, the actual phases have $p \in (2^{-d-1}, 2^{-d}]$, so a suitable fixed ε covers the entire dyadic block. For $d = 0$, the complete-layer edge $p \rightarrow 1$ must instead be treated by the exact deletion formula (12). This is a genuine boundary change of asymptotic regime, not an error to be removed by formally substituting $p = 1$ in the Gaussian prefactor.

9.1 The fixed-deletion regime

For fixed $r \geq 0$ and $N = 2^m$, Corollary 4.3 gives

$$b_{N-r}^{(0)} = u_m \frac{N^r}{r!} \left[1 - \frac{rm + r(r-1)/2}{N} + O_r \left(\frac{m^2}{N^2} \right) \right]. \quad (37)$$

This follows directly by expanding $\binom{N-m}{r} = N^r/r! \prod_{i=0}^{r-1} (1 - (m+i)/N)$. More generally, for $r = o(N)$ with $m+r < N$, an exact logarithmic series is

$$\log \frac{r! b_{N-r}^{(0)}}{u_m N^r} = - \sum_{\ell \geq 1} \frac{1}{\ell N^\ell} \sum_{i=0}^{r-1} (m+i)^\ell. \quad (38)$$

It converges because $(m+r-1)/N < 1$, and truncation is controlled by the geometric tail of the logarithm series. This supplies a separate exact route toward a matched sparse-to-bulk expansion.

10 Arc counts and a binomial law near complete layers

10.1 A core-dependent polynomial with a simple complement form

Let $e(F)$ be the number of arcs in the membership graph of F . Since arcs are membership incidences,

$$e(F) = \sum_{S \in F} |S|.$$

For $n = 2^m - r$ with $0 \leq r < 2^{m-1}$, an extremizer is $F = \mathcal{P}(C) \setminus R$, where $C \in \mathcal{U}_m$ and $R \subseteq \mathcal{P}(C) \setminus C$ has size r . Hence

$$e(F) = m2^{m-1} - Z, \quad Z = \sum_{S \in R} |S|. \quad (39)$$

In particular, *all* extremizers at $n = 2^m$ have exactly $m2^{m-1}$ arcs, regardless of the core graph.

For a fixed core C , the full deficit-counting polynomial is

$$[z^r] \prod_{S \in \mathcal{P}(C) \setminus C} (1 + zw^{|S|}). \quad (40)$$

Summing this over $C \in \mathcal{U}_m$ gives the exact unlabeled joint count by size and arc deficit. Equivalently, the product is the formal quotient

$$\frac{\prod_{j=0}^m (1 + zw^j)^{\binom{m}{j}}}{\prod_{x \in C} (1 + zw^{|x|})}.$$

The denominator divides the numerator as a polynomial product; it is not an analytic singularity assertion.

Theorem 10.1 (Total-variation binomial approximation). *Let F be uniform among extremal EAD isomorphism classes on $n = 2^m - r$ vertices, where $m \geq 1$ and $0 \leq r < 2^{m-1}$. Define $Z = m2^{m-1} - e(F)$. Then*

$$d_{\text{TV}}(\mathcal{L}(Z), \text{Bin}(rm, 1/2)) \leq \min \left\{ 1, \frac{rm + \binom{r}{2}}{2^m} \right\}. \quad (41)$$

The same statement holds for a uniformly labeled extremizer.

Proof. By Theorem 4.2, the core is uniform on $\mathcal{U}_m$, and, conditional on it, R is a uniform r -subset of the available family. Fix a core and draw r independent uniform subsets $S_1, \dots, S_r$ of the entire Boolean cube $\mathcal{P}(C)$. Each S_i has a $\text{Bin}(m, 1/2)$ size, independently; therefore $\sum_i |S_i|$ is exactly $\text{Bin}(rm, 1/2)$.

Condition this ordered sample on avoiding the m forbidden core subsets and on having no repetition. The conditional distribution is exactly uniform sampling without replacement from $\mathcal{P}(C) \setminus C$, with an irrelevant ordering. By the union bound, the conditioning event fails with probability at most

$$r \frac{m}{2^m} + \binom{r}{2} \frac{1}{2^m}.$$

Conditioning changes total variation by the failure probability, and mapping to the sum of the sizes cannot increase total variation. The estimate is independent of the core, so it survives mixing over cores. Rigidity implies that uniform labeling gives the same induced law on isomorphism classes, proving the last assertion. $\square$

Corollary 10.2 (Boundary central limit law). *If $m \rightarrow \infty$, $r = r(m) \geq 1$, and $r(m+r) = o(2^m)$, then*

$$\frac{m2^{m-1} - e(F) - rm/2}{\sqrt{rm}/2} \implies N(0, 1). \quad (42)$$

Proof. The total-variation error in (41) tends to zero. For $B \sim \text{Bin}(rm, 1/2)$, the characteristic function of $(B - rm/2)/(\sqrt{rm}/2)$ is $\cos(t/\sqrt{rm})^{rm}$, which tends to $e^{-t^2/2}$. The claimed convergence follows, since $rm \rightarrow \infty$. $\square$

For fixed $r \geq 1$, the error is $O_r(m2^{-m})$ and the Gaussian limit already applies. For $r = 0$, the distribution is instead deterministic. The theorem concerns graphs conditioned to maximize their source count; it is not Wagner’s normal limit for unrestricted EADs.

10.2 Exact conditional moments

Let $N = 2^m$, $M = N - m$, and define two core statistics

$$e_C = \sum_{x \in C} |x|, \quad h_C = \sum_{x \in C} |x|^2.$$

The sum and squared sum of subset sizes in the available population are

$$A_1 = \frac{mN}{2} - e_C, \quad A_2 = \frac{(m^2 + m)N}{4} - h_C.$$

Writing $\mu_C = A_1/M$ and $v_C = A_2/M - \mu_C^2$, uniform sampling without replacement gives

$$\mathbb{E}(Z \mid C) = r\mu_C, \quad \text{Var}(Z \mid C) = \frac{r(M-r)}{M-1}v_C \quad (M > 1). \quad (43)$$

The variance identity follows by expanding the variance of the sum of r sampled population entries: the covariance of two distinct sampled entries is $-v_C/(M-1)$. This derivation also specifies the variance normalization. When $M = 1$, the sample is deterministic. These exact moments remain available outside the small-error range of Theorem 10.1.

11 Exact verification and computational artifacts

11.1 Three complementary checks

The accompanying program performs the following distinct checks. First, it computes u_n from (15) and then evaluates the marked-source sieve. Second, it computes the source triangle using the source-deletion recurrence from Tomescu’s Corollary 2.1.7 [7]:

$$\begin{aligned} u(n, s) = \frac{1}{s} & \left[(2^{n-s} - (n-1))u(n-1, s-1) \right. \\ & \left. + \sum_{j=0}^{n-s-1} \binom{s+j}{j+1} 2^{n-1-s-j} u(n-1, s+j) \right]. \end{aligned} \quad (44)$$

Here $u(1, 1) = 1$ and out-of-range entries are zero. The code verifies exact divisibility by s , equality of the two triangle constructions, the complete support theorem, the finite-defect formulas, and the covering inequalities. [Added 5 October 2026, batch 98: the recurrence (44) is quoted here from Tomescu without proof; both Part II manuscripts give a direct bijective proof, printed as Proposition 15.4.]

Third, for every $n \leq 7$, the program enumerates all graphs in which arcs point from a larger vertex index to a smaller one. It rejects repeated out-neighborhoods, collapses surviving graphs to nested finite sets, and deduplicates the resulting isomorphism classes. This does not use either counting recurrence. Source counts and arc counts are computed directly from the arc masks. Every DAG has such an ordering, so the enumeration covers every class, even though some classes occur under several orderings.

Table 1: Exhaustive enumeration, compared with the recurrence counts. The middle column counts surviving graphs in a fixed topological labeling convention, before isomorphism deduplication.

n	Surviving ordered graphs	Classes u_n	Positive source row $u(n, k)$
1	1	1	1
2	1	1	1
3	2	2	2
4	10	9	8, 1
5	120	88	68, 20
6	3240	1802	1248, 534, 20
7	187920	75598	48640, 25008, 1940, 10

11.2 Recorded results

The recorded run used `--max-n 64 --brute-n 7`. It checked 2080 triangle cells by both recurrences, 2080 covering inequalities, 64 extremizer counts, and 354 finite-defect entries with $d \leq 5$. It also matched the first 17 displayed terms of A001192 and the first 25 displayed flattened labeled entries of A182162 [3, 2]. For each $n \leq 7$, the directly enumerated extremal arc polynomial agreed with the Boolean-core construction.

Table 2: Exact unlabeled extremizer counts. Labeled counts are obtained by multiplying the final column by $n!$.

n	q	$a(n)$	$b_n^{(0)}$	n	q	$a(n)$	$b_n^{(0)}$
1	0	1	1	9	4	5	7128
2	1	1	1	10	4	6	8316
3	2	1	2	11	4	7	7128
4	2	2	1	12	4	8	4455
5	3	2	20	13	4	9	1980
6	3	3	20	14	4	10	594
7	3	4	10	15	4	11	108
8	3	5	2	16	4	12	9

The source-count CSV files contain exact integers, not rounded plots or floating-point approximations. A separate `check_asymptotics.py` uses 70-digit Decimal arithmetic to test the first correction at $n = 192, 768, 3072, 12288$ and $d = 0, 1, 2$; all twelve tests improve the leading approximation. Its separate CSV records numerical diagnostics, not interval-certified bounds. The JSON report records the checks and their ranges. No finite experiment is presented as a proof of non-P-recursiveness, the entropy theorem, or a limit distribution.

11.3 Computational complexity and numerical stability

For fixed d , (16) uses only $d + 1$ terms after the total counts through $q + d$ have been tabulated. This is an arithmetic-operation statement, not a bit-complexity claim: the binomial coefficients and the answers themselves have $\Theta_d(n)$ bits in the bulk. Near a complete layer, the exact complement form (12) can be much smaller.

Use integer binomial coefficients for exact evaluation. For large floating-point evaluations, use logarithms and either the covering approximation or the all-orders expansion; do not subtract enormous, nearly equal floating-point source-sieve terms near small parameters. The

program caps brute-force enumeration at seven vertices to make the exponential nature of that check explicit rather than silently attempting an infeasible extension.

[write] **Added 3 October 2026, batch 85.** In the repository the two programs ship as `code/verify.py` and `code/check_asymptotics.py`, and their outputs under `data/`. The delivered `data/verification-output.txt` is not shipped: it was a byte copy of `data/verification.json`, since the program prints the JSON report it writes. Both programs write beside themselves (into a `data` directory next to the script, the second without an option), so the report's README gives a rerun on a scratch copy in the delivered layout. The intake ran them so on Python 3.14.4: `verify.py --max-n 64 --brute-n 7` passed in about 7 s, `check_asymptotics.py` in about 4 s; the three CSV files were reproduced byte for byte, the JSON report up to Windows line endings, and both printed transcripts matched the recorded ones. The 17 terms of A001192 and 25 terms of A182162 that `verify.py` embeds as test fixtures are OEIS data, published under CC BY-SA 4.0.

12 Formalization route and further research

12.1 A finite, auditable formalization route

A formal development need not begin with the full set-theoretic collapse. Represent a finite digraph by a Boolean relation on a finite vertex type. Define the finite set of out-neighbors, sourcehood, acyclicity, and extensionality. The maximum-source proof then reduces to an injection from vertices into the finite powerset of the nonsources, followed by the explicit construction in Theorem 3.1.

A useful next layer would prove that deletion of sources preserves out-neighborhoods and extensionality. This supports a marked-source bijection and finite binomial inversion. To treat unlabeled counts, one may either prove rigidity and use labeled counts throughout, or introduce finite collapse and quotient by graph isomorphism. The former avoids set-theoretic infrastructure for the first main counting results.

The entropy and non-holonomicity proofs can be separated from graph semantics. A generic exponential-jump lemma is particularly suitable for reuse: its hypotheses are scalar asymptotic statements, and its proof only needs eventual polynomial-ratio bounds and the triangle inequality. The deliverable contains no Lean source and makes no claim of kernel verification. This paragraph is a development plan, not an assertion that particular library declarations already implement the argument.

12.2 Specific research questions

The following questions are proposed directions, not claims that each has been certified open in the published literature.

1. Natural boundaries of the extremizer generating functions. Does $B_d(z)$ have a natural boundary on $|z| = e^{-\lambda_{d+1}}$? The proof here establishes non-D-finiteness but not a dense set of singularities. Unlike the lacunary maximum sequence, B_d has positive coefficients throughout its eventual support. A useful first target is an analysis of contributions from short intervals immediately after powers of two.

[write] **Added 5 October 2026, batch 98: answered.** Yes, for every fixed $d \geq 0$: Part II proves that $|z| = R_d$ is a natural boundary of B_d (Theorem 14.2). Two independent manuscripts of batch 98 prove it, both by the mechanism suggested here: on each dyadic block the coefficients just after the power of two, normalized, approach a geometric profile uniformly, and at roots of unity of power-of-two order the late blocks start in phase, so B_d blows up along a dense set of radii. Part II adds the radial phase with an $O(t)$ rate, the two-term radial growth law, an L^p comparison on every

arc, natural boundaries for the arithmetic sections of the coefficient sequence (of power-of-two step in one manuscript, of every step in the writing step's Theorem 21.5), and the absence of a bidisk of convergence for the size-and-defect series. The proofs are unrefereed and not formalized.

2. A sharper small-core amplitude. Combine an explicit, verified asymptotic expansion for u_m with (35). Since $m \asymp \log n$, the resulting corrections live on scales different from the Boolean-layer saddle. Distinguish statements uniform in the dyadic phase from fixed-phase expansions.

3. Defects growing with the graph size. How far can the entropy law and exponential-jump obstruction be extended when $d = d(n)$ increases? The covering theorem itself remains valid, but $\log u_{q+d}$ ceases to be a negligible amplitude if the core becomes too large. Identify the transition at which the large-source boundary analysis must connect with unrestricted EAD enumeration.

4. A matched bulk-to-complete-layer expansion. The bulk formula and the fixed-deletion formula overlap when the number $r = 2^m - n$ tends to infinity slowly. Derive explicit uniform remainder bounds covering r from fixed values to a positive fraction of 2^m , without using a Gaussian prefactor at a degenerate endpoint.

5. Extend the arc law beyond the collision estimate. The total-variation coupling loses strength when r is comparable to $2^{m/2}$. This is a limitation of comparing whole subset samples, not necessarily of comparing their sizes. Can a finite-population local limit theorem recover the arc distribution for all densities, with an explicit correction for the m forbidden core subsets?

6. Joint degree statistics. Study the empirical outdegree profile of source vertices and the joint indegree distribution of core vertices. The source-neighborhood family is a sample without replacement from a Boolean cube with a small deleted family. This suggests multivariate hypergeometric limits, but the exact core-induced correlations should be retained rather than discarded.

7. Source layers under repeated deletion. Successively delete all sources. Conditional on the first layer being extremal, the remaining graph is exactly a uniform small core. Can this be iterated to obtain precise distributions for the number of rounds, the full layer-size vector, or the maximum rank under extremal source conditioning?

8. Restricted degree and weak extensionality. If outdegrees are at most Δ , the Boolean capacity becomes $\sum_{j \leq \Delta} \binom{m}{j}$ rather than 2^m . Determine when its pigeonhole bound is sharp, how covering changes, and whether analogous boundary jumps imply non-P-recursiveness. A related problem permits bounded multiplicity of out-neighborhoods or distinguished atoms.

9. Exact ranking and sampling of boundary classes. Implement ranking and unranking for the core-plus-subset representation. A ranked core together with a combinatorial-number-system rank for its chosen neighborhoods gives a natural interface. Specify bit complexity, including the cost of representing deeply nested finite sets.

10. Formal proofs and OEIS editorial updates. Formalize the source support theorem and the boundary bijection first. After independent mathematical review, the proof of A182220 and the boundary formula can be proposed to OEIS with the historical upper-bound citation. No OEIS submission or repository commit was made during this work.

[write] **Added 5 October 2026, batch 98.** Question 1 is answered in Part II (note above). The further questions of Part II (Section 24) take up Questions 3 (growing defects), 6 (degree and arc statistics), 8 (restricted outdegree) and 10 (formalization) again from the analytic side; they remain open as stated here.

13 Conclusion

The source maximum has a short Boolean-capacity proof, and its known upper bound should not be confused with a new breakthrough. The interesting structure appears when one counts the graphs attaining, or nearly attaining, that maximum. Their core has logarithmic size, while their sources select a large family of subsets of that core. At maximum density, coverage is automatic and the count is exactly $u_q \binom{2^q - q}{n - q}$. At fixed defect, only finitely many terms of the classical sieve survive.

This decomposition makes it possible to prove a uniform entropy law without solving the separate asymptotic problem for the core count. Its phase resets at powers of two. The resulting exponential jumps rule out every polynomial-coefficient linear recurrence for each fixed boundary diagonal. Exact arc formulas and the binomial coupling provide further information about the extremizers themselves.

The principal deliverable is therefore a chain of proved results: sharp source support, exact boundary enumeration, finite-defect formulas, uniform high-source estimates, complete growth-rate intervals, and non-holonomicity, with reproducible finite checks. Questions about historical priority, natural boundaries of the extremizer series, growing defects, and full formal verification remain explicitly separate.

[write] **Added 5 October 2026, batch 98.** Of these, natural boundaries of the extremizer series are no longer open: Part II proves that $|z| = R_d$ is a natural boundary of B_d for every fixed d . Priority, growing defects and formal verification remain separate, for Part II as well.

Part II

Natural boundaries at the source edge and at the source extremum

Summary. Let $u(n, s)$ count isomorphism classes of extensional acyclic digraphs with n vertices and s sources, and for a fixed defect $d \geq 0$ let $b_n^{(d)} = u(n, n - \lceil \log_2 n \rceil - d)$, with inadmissible terms zero, and $B_d(z) = \sum_{n \geq 1} b_n^{(d)} z^n$, as in Part I. We prove that the circle of convergence

$$|z| = R_d, \quad R_d = \frac{(c_d - 1)^{c_d - 1}}{c_d^{c_d}}, \quad c_d = 2^{d+1},$$

is a natural boundary of B_d for every fixed d . This answers Question 1 of Section 12.2. The proof gives more than noncontinuation. At every root of unity ζ of power-of-two order, the quotient $B_d(R_d e^{-t} \zeta) / B_d(R_d e^{-t})$ has an explicit nonzero limit, with error $O_{d, \zeta}(t)$. Writing $L = \log(1/t)$, the logarithm of the radial magnitude is $L^2 / (2 \log 2) + L \log L / \log 2 + O_d(L)$. On every nonempty open arc, every fixed positive power of the modulus has integral comparable to the positive radial value raised to that power, so the boundary is strong in a quantitative local L^p sense. Every arithmetic subsequence of power-of-two step also has a natural boundary, and the joint size-and-defect series has no nontrivial bidisk of convergence.

The mechanism is a uniform geometric limit for the normalized coefficients of each dyadic block, together with the common phase of the late block starts at dyadic roots of unity. All graph-counting estimates and analytic arguments are proved here; the counting identities inherited from Part I and from the classical literature are credited and re-proved, not claimed. Exact computations and numerical diagnostics accompany the proofs; the analytic conclusions do not rely on finite experiments. These are conventional proofs in unrefereed manuscripts, not formally verified, and not certified as historically first.

[write] **Added 5 October 2026, batch 98: provenance of Part II.** Part II is a merge of two manuscripts, manuscripts 01 and 05 of batch 98 of ProveIt’s incoming-reports intake. Both arrived in commit 2172df76a and were placed in this report by commit b50febf79 (batch 98C). Both pin the repository at commit d18416ec7e187a0948248cb3077e37d7b089a8f9, at which this report consisted of the text printed here as Part I, without the batch-98 notes (it was unchanged from the pin to the placement); both answer its Question 1, independently of each other.

	source 05 (<i>base</i>)	source 01
archive	OEIS_Source_Boundaries.zip, wrapper ois_natural_boundaries/, main file article.tex (1,462 lines)	OEIS_EAD_Natural_Boundaries.zip, wrapper ead-natural-boundary/, main file ead_natural_boundaries.tex (1,571 lines)
title	<i>Natural Boundaries at the Source Extremum: Fixed deficits, dyadic radial limits, and local L^p growth</i>	<i>Natural Boundaries at the Source Edge of Extensional Acyclic Digraphs: Dense singularities, universal radial limits, and logarithmic growth for OEIS source-count diagonals</i>
date, author field	5 October 2026 (UTC); “Research manuscript prepared with ChatGPT”, PDF author “ChatGPT”	5 October 2026; PDF author “Research prepared with ChatGPT”
shipped as	prefix 03-source-extremum-; labels sbd:xtm:	prefix 02-source-edge-; labels sbd:edge:
contributes	every shared theorem in its stronger form: the $O(t)$ radial phase, the two-sided local L^p comparison, the block profile on a disk larger than the circle; concentration of block mass; the no-bidisk theorem; first block and amplitude corrections	the abstract transfer lemma for coherent dyadic blocks; natural boundaries of all power-of-two arithmetic sections, with their radial ratios; second proofs of the radial law, the growth law and the log-level arc law; a direct proof of the deletion recurrence

The two texts are not editions of one another: their shared word 8-grams are 2.7% of manuscript 01's and 2.3% of manuscript 05's, and their notation, proofs and extra results differ. The local file prefixes continue this report's sequence (the unprefixes Part I counts as 01) in archive order. Neither manuscript's L^AT_EX source, PDF or delivery README is shipped; they survive in the archives of the arrival commit, and the report's README says how to retrieve them.

Where the merge chose. Manuscript 05 is the base: it has the same hypotheses as 01 (every fixed $d \geq 0$) and is equal or stronger on every shared theorem. Theorems proved by both are printed once, in 05's form, and credited to both; where 01's proof differs it is printed as a marked second route. Everything only in 01 is printed in full: the transfer Lemma 18.1, the arithmetic-section Theorem 21.1 with its example and corollary, and the direct proof of the deletion recurrence. Statements that both manuscripts re-prove from Part I are printed as pointers (Section 15), except where a proof differs or is new to the report. The two further-question lists are merged with each other and with Part I's (Section 24); the two proof-status ledgers are merged in Section 25. Nothing was selected away.

Overlap with Part I, credited. Both manuscripts re-prove, for self-containment, the finite collapse and rigidity (Lemma 2.1, Corollary 2.2), the source support (Theorem 3.1), the intrinsic core and automatic covering at $d = 0$ ((8), Lemma 4.1, Theorem 4.2), the upper bound of Lemma 2.3, the marked-source sieve (Proposition 5.1, Corollary 5.2), the finite-defect formula (Theorem 5.3), the covering estimate (Theorem 6.1), the radius (34) and the non-D-finiteness of Theorem 8.2. Only the radius and non-D-finiteness are re-derived by a new route (from the block profile and the natural boundary). The chain lower bound for u_m and the bijective proof of the deletion recurrence (44) are new to the report. Neither manuscript uses a theorem of Part I as a premise.

Editorial changes. The manuscripts' references to "the starting report", "the source report" or "the ProveIt report" (their citations `baseline` and `repo-report`, whose pin and path this note records) now point to Part I. Their sections are renumbered and interleaved as Sections 14–25; manuscript 05's two appendices are Sections 23 and 25; manuscript 01's Appendices A and B are in Sections 15 and 25, and its Appendix C (rebuilding its package) is replaced by the report's README. The symbols of manuscript 01 are translated into those of manuscript 05 (table below), and u_m replaces 05's U_m . The bibliographies are merged into the report's. The shipped table files are input under their prefixed names, through a wrapper that adds the label sub-prefix to the names they define and cite; the figures are the shipped prefixed files. Manuscript 05 wrapped every table input and figure in `\IfFileExists`, which silently drops a missing file; here they are input unconditionally, so the build fails if one is missing. Manuscript 05's inline table of small counts duplicates Part I's table in Section 11.1 and manuscript 01's shipped table, and is not printed again. Statements marked [write] are the writing step's: two results with proofs (Lemma 21.4 with Theorem 21.5, and Proposition 24.1), the notes, and the merged question list. Corollary 21.7 and the paragraphs headed "Dated note (5 October 2026)" in Sections 21.2, 22 and 24 were added by the intake after its check of the write. No statement, proof or number of either manuscript was otherwise altered.

Status and non-claims, kept from both sources. Both manuscripts are unrefereed and prepared with ChatGPT; nothing in Part II is formalized in Lean or Rocq, and its place in this report confers no formal status. Their limited literature searches certify no historical priority. The counting identities, collapse, rigidity, sieve, covering bound and source maximum are background (Tomescu's thesis [7], Johnston's 2012 sieve in A182162 [2], Policriti–Tomescu [6], Part I), re-proved, not new. Wagner's results [8, 10] are context, never premises (manuscript 01 could not access the full text of the journal version and read its abstract). Breuer and Simon [11] supply terminology only; their bounded-coefficient theorem is not applied. Floating-point diagnostics are not interval-certified (truncation is bounded analytically, roundoff is not enclosed), and finite checks are tests, not proofs. Nothing is claimed about radial behaviour at non-dyadic angles or at almost every angle, about nonlinear differential algebraicity, or about complex defect fugacity. Manuscript 05's statement that "independent internal reviews" checked its proofs is reported as its own description of its preparation; the intake could not verify it, and it is not evidence of review (see Section 24). The manuscripts made no repository commit, OEIS submission or message to another person.

[write] **Added 5 October 2026, batch 98: notation of Part II.** Part II writes everything in manuscript 05's letters. The middle column gives manuscript 01's symbol for the same object; the right column gives the meaning the letter has in Part I, the tempting false reading. Shared notions keep Part I's symbols: $u(n, s)$, u_m (A001192; manuscript 05 writes U_m), $q = q(n) = \lceil \log_2 n \rceil$, $b_n^{(d)}$, B_d , R_d , $(n)_m$.

symbol	meaning in Part II	in source 01	not: Part I's meaning
$c = c_d$	2^{d+1}	a	(c a core element)
$a = a_d$	$c - 1 = 2^{d+1} - 1$	$a - 1$	$a(n) = \text{A182220}(n)$
$R = R_d$	$a^a/c^c = e^{-\lambda_{d+1}}$, the radius (34)	R_d	R a removed family
aR	$(1 - 1/c)^c < e^{-1}$	ρ_d	$\rho_n = n/2^q$, the dyadic phase
h	$\log 2$	$\log 2$; also the step 2^s	the order of a recurrence (32)
$K = K_q$	2^{q-1} ; block $K < n \leq 2K$	N	($N = 2^m$ in Part I)
m, k_0	$q + d$ (so $2^m = cK$); $k_0 = K + 1 - m$	$m; N + 1 - m$	m the core size (same)
C_q	$\binom{cK-m}{k_0}$	$C_q(N + 1)/u_m$	C a core; $C_1(m, p)$
A_q	$u_m C_q R^{K+1}$, the block amplitude	c_q	$A = u_m \binom{M}{k}$ (Theorem 6.1)
$u_m C_q$	the positive leading count of block q	A_q	—
$P_q(z)$	block polynomial (60) in z	$P_q(w) = P_q(Rw)$	$P_n(z) = \sum_s u(n, s)z^s$
$g(z)$	$1/(1 - az)$; $g(Rw)$ is 01's profile	$1/(1 - \rho w)$	—
E_d	the head polynomial; $E_{c,m}$ an amplitude coefficient	$E_d(w)$	$E(n, a(n)); \mathbb{E}$ expectation
t, L	radial variable, $L = \log(1/t)$	$\varepsilon; r = e^{-\varepsilon}; L$	t marked sources; L a source family, $L(z)$
$\mathcal{T}(L)$	$L^2/(2h) + L \log L/h$	$T(L)$	$T = 2^m; T(n, k)$
T	a real radius, $R < T < 1/a$	—	$T = 2^m$
$S_d(t)$	$\sum_q A_q e^{-t(K_q+1)}$	$S(r), S_\varepsilon$	$S_\ell(m)$ power sums
p	a positive exponent (L^p)	p	$p = n/T$ a phase
M	a positive power in (51)	M	$M = 2^m - m$
D, D_t, F_t	an integer offset; $\sum n b_n R^n e^{-tn}$; $B_d(R e^{-t})$	—	—
γ_d	coefficients of a finite combination $\sum \gamma_d B_d$	c_d	$\lambda_d = 2^d H(2^{-d})$ (05 wrote λ_d)
r, r_0	a step $r \geq 1$ and a residue $0 \leq r_0 < r$	$h = 2^s, r_0$	$r = 2^m - n$
$B_d^{(r, r_0)}, G_d^{(r, r_0)}$	the residue-class series (101)	$H_{d, h, r_0}, G_{d, h, r_0}$	H the binary entropy
α_q, Π_q, Π	amplitudes and polynomials of the abstract Lemma 18.1	c_q, P_q, P	$\alpha < \beta$ in Lemma 8.1
$\omega_q(t), y_0(L), q_t, \varphi$	the write's block weights and phase, Section 21.2	—	Φ_d the entropy profile
$\mathcal{B}(z, v)$	the size-and-defect series (98)	—	—

Manuscript 01 used h both for $\log 2$ (its Lemma 7.1) and for the step 2^s (its Section 9); here h is always $\log 2$. It used L both for a source family and for $\log(1/\varepsilon)$, and $L_q = N + 1$ for a block start; here L is only $\log(1/t)$ and the block start is $K + 1$. Its set $\mathcal{D}$ of dyadic roots of unity is written out in words. Within proofs, $C, C_d, \kappa, \eta, \epsilon$ and δ denote local constants.

14 The question and the results

14.1 From a source maximum to the number of near-extremizers

An extensional acyclic digraph is a finite directed graph with no directed cycle in which different vertices have different out-neighborhoods; these graphs encode finite transitive sets, an arc from x to y corresponding to $y \in x$. OEIS A001192 counts their isomorphism classes, and A182162 refines their labeled counts by the number of sources [3, 2]. The relevant OEIS entries distinguish the following quantities [1, 2, 3, 4].

Entry	Quantity
A001192	u_m , the number of finite transitive sets of size m , equivalently EAD isomorphism classes.
A182161	$m!u_m$, the number of EADs on a fixed labeled vertex set.
A182162	$n!u(n, k)$, the labeled source-count triangle.
A182220	$n - \lceil \log_2 n \rceil$, the maximum possible number of sources.

The source maximum is

$$s_{\max}(n) = n - q(n), \quad q(n) = \lceil \log_2 n \rceil \quad (n \geq 1), \quad (45)$$

the quantity in A182220 [1]. Its elementary upper bound is in Tomescu's thesis, printed page 29 [7]; Part I supplies the matching construction and develops the enumeration near this boundary (Theorem 3.1).

For a fixed integer $d \geq 0$, as in (2) and (33),

$$b_n^{(d)} = \begin{cases} u(n, n - q(n) - d), & n - q(n) - d \geq 1, \\ 0, & n - q(n) - d < 1, \end{cases} \quad B_d(z) = \sum_{n \geq 1} b_n^{(d)} z^n. \quad (46)$$

Thus $d = 0$ counts the source-maximizing graphs, and $d = 1$ counts graphs with one fewer source. A fixed d leaves a core of $q(n) + d$ nonsources, which is only logarithmic in n . When subsequences are extracted we also set $b_0^{(d)} = 0$, and $\binom{M}{k} = 0$ for $k < 0$ or $k > M$.

Part I proves that each B_d is not D-finite and has radius R_d (Theorem 8.2), and asks, in Section 12.2, Question 1, whether its entire circle of convergence is a natural boundary; it distinguishes that question expressly from non-D-finiteness (Remark 8.3). Non-D-finiteness by itself would not answer it. The argument below identifies a dense set of radial singularities and quantifies their size.

14.2 Conventions and main theorems

All logarithms without a subscript are natural. A statement with an O_d remainder permits its constant to depend on d ; it does not assert uniformity for a growing defect. For fixed d , set

$$c = c_d = 2^{d+1}, \quad a = a_d = c - 1, \quad R = R_d = \frac{a^a}{c^c}, \quad h = \log 2. \quad (47)$$

Then $R_d = e^{-\lambda_{d+1}}$ is the radius (34) of Part I, since $cH(1/c) = c \log c - a \log a$, and $0 < aR = (1 - 1/c)^c < 1$. A *dyadic root of unity* is a complex number ζ with $\zeta^{2^j} = 1$ for some integer $j \geq 0$; these form a dense subset of the unit circle containing 1.

Definition 14.1 (source 01). The circle $|z| = R$ is a *natural boundary* for a holomorphic function on $|z| < R$ if the function has no holomorphic continuation through any point of that circle. Continuation through a point means a holomorphic function on a neighborhood of the point that agrees with the original function on the portion inside its disk.

Theorem 14.2 (Natural boundary and quantitative radial phase; sources 05 and 01). *For every fixed integer $d \geq 0$, B_d has radius of convergence R_d . The circle $|z| = R_d$ is a natural boundary. Moreover, for every fixed dyadic root of unity ζ ,*

$$\frac{B_d(R_d e^{-t} \zeta)}{B_d(R_d e^{-t})} = \zeta \frac{1 - a_d R_d}{1 - a_d R_d \zeta} + O_{d, \zeta}(t) \quad (t \downarrow 0). \quad (48)$$

The displayed limit is nonzero, and $B_d(R_d e^{-t}) \rightarrow +\infty$.

In manuscript 01's normalization ($\rho_d = a_d R_d$, $r = e^{-t}$) the limit reads

$$\lim_{r \uparrow 1} \frac{B_d(R_d r \zeta)}{B_d(R_d r)} = \frac{\zeta(1 - a_d R_d)}{1 - a_d R_d \zeta}. \quad (49)$$

Manuscript 05 proves the theorem with the $O(t)$ rate (Section 18); manuscript 01 proves the limit (49) and the natural boundary through the abstract transfer Lemma 18.1 (Section 18.4).

Theorem 14.3 (Radial size; sources 05 and 01). *For each fixed $d \geq 0$, with $L = \log(1/t)$ and $\mathcal{T}(L) = L^2/(2h) + L \log L/h$,*

$$\log B_d(R_d e^{-t}) = \frac{L^2}{2h} + \frac{L \log L}{h} + O_d(L). \quad (50)$$

At every fixed dyadic root ζ , the same formula holds for $\log |B_d(R_d e^{-t} \zeta)|$, with the constant in the error allowed to depend on ζ . In particular, for every fixed $M > 0$,

$$t^M |B_d(R_d e^{-t} \zeta)| \rightarrow \infty. \quad (51)$$

Theorem 14.4 (Local L^p comparison and growth; source 05, the log-level law also source 01). *Let $d \geq 0$ and $p > 0$ be fixed, and let I be a nonempty bounded open interval of angles. For all sufficiently small $t > 0$,*

$$C_{d,p,I}^{-1} B_d(R_d e^{-t})^p \leq \int_I |B_d(R_d e^{-t+i\theta})|^p d\theta \leq C_{d,p,I} B_d(R_d e^{-t})^p \quad (52)$$

for a positive constant $C_{d,p,I}$. Consequently,

$$\log \int_I |B_d(R_d e^{-t+i\theta})|^p d\theta = \frac{pL^2}{2h} + \frac{pL \log L}{h} + O_{d,p,I}(L). \quad (53)$$

In particular these radial integral means tend to infinity on every nonempty open arc, for every $p > 0$.

The logarithmic law (53) is manuscript 01's Theorem 1.4, proved there by a different, single-point argument (Section 19); the two-sided comparison (52) is manuscript 05's.

Section 21 proves a further stability result (manuscript 01): for every $s \geq 0$ and $0 \leq r_0 < 2^s$ the series $\sum_{\ell \geq 0} b_{2^s \ell + r_0}^{(d)} z^\ell$ has natural boundary $|z| = R_d^{2^s}$; the convention at $n = 0$ does not affect this. The writing step extends it to every step $r \geq 1$ (Theorem 21.5). Section 20 establishes a zero-radius result for positive defect fugacity (manuscript 05).

14.3 Normalization and the scope of the contribution

Extensional acyclic digraphs have no nontrivial automorphisms (Corollary 2.2). Hence the labeled count corresponding to $b_n^{(d)}$ is $n! b_n^{(d)}$, and B_d is both the *ordinary* generating function of the unlabeled counts and the *exponential* generating function of the corresponding A182162 entries. The ordinary generating function of the labeled counts has radius zero; it is not the analytic function of Theorem 14.2.

Finite transitive sets and their enumeration are classical [5, 6, 7]; Wagner's work treats unrestricted asymptotic enumeration and several graph statistics [8, 10]. These sources concern the background. The analytic conclusions follow from the elementary estimates proved below and require no imported precise asymptotic formula for u_m . Natural-boundary arguments using lacunary exponents have a long history; Breuer and Simon discuss both classical and strong natural boundaries [11]. Manuscript 05 uses their terminology, not their bounded-coefficient right-limit theorem: the normalized coefficients here are unbounded. The contribution is a proved application to this precise enumeration problem, with quantitative refinements. Limited primary-source searches by both manuscripts located no earlier treatment of these fixed-defect natural boundaries; that does not establish worldwide priority.

Table 3: What Part II inherits and what it proves (after manuscript 05’s table; “starting report” is Part I).

Statement	Status in Part II
Source maximum and Boolean-core representation	In Part I; re-proved with attribution.
Marked-source sieve and core-count recurrence	Classical enumerative input; direct proofs included.
Fixed-defect radius and non-D-finiteness	Already in Part I; recovered by a different analytic argument.
Natural boundary for every fixed defect	Resolves Part I’s Question 1.
Radial phase with $O(t)$ error; two-term logarithmic growth	Additional quantitative theorems.
Local L^p comparison on every arc	Additional strong-boundary theorem.
Power-of-two arithmetic sections	Natural boundaries, with explicit radial ratios (manuscript 01).
No bidisk for the size-and-defect series	Additional obstruction to a holomorphic joint generating function.

15 Finite graphs, finite sets and source cores: what Part II takes from Part I

Both manuscripts begin by re-proving the combinatorial background, to make their analytic conclusions independently assessable. These statements are Part I’s and are not printed again; the table records the correspondence (the middle columns use the manuscripts’ own numbering) and Remark 15.1 the proofs that differ.

statement	source 05	source 01	Part I
finite collapse, rigidity, $n!u(n, s)$ labeled graphs	Lemma 2.1, Corollary 2.2	Lemma 2.1	Lemma 2.1, Corollary 2.2
source support, $s_{\max}(n) = n - q(n)$ intrinsic core $C = \bigcup F$, $F = C \cup L$, covering condition	Proposition 2.3 §2.3	Proposition 2.2 §2.1	Theorem 3.1 (8), (9)
$u_m \leq 2^{\binom{m}{2}}$	Lemma 2.4	Lemma 4.1	Lemma 2.3
marked sources, inversion, total recurrence	Proposition 3.1	Proposition 3.1	Proposition 5.1, Corollary 5.2
finite-defect formula, $b_n^{(0)} = u_q\left(\frac{2^q - q}{n - q}\right)$	Corollary 3.2	Corollary 3.2; display after Lemma 2.3	Theorem 5.3, Theorem 4.2
covering estimate $0 \leq u_m\left(\frac{2^m - m}{k}\right) - u(m + k, k) \leq$ $m2^{-k}u_m\left(\frac{2^m - m}{k}\right)$	Lemma 3.3	Lemma 2.3	Theorem 6.1
covering automatic at $d = 0$ deletion recurrence	after Lemma 3.3 §9.1 (sketch)	after Lemma 2.3 Appendix A	Lemma 4.1 (44), quoted

In particular the covering estimate is used below in the form

$$0 \leq u_m\left(\frac{2^m - m}{k}\right) - u(m + k, k) \leq m2^{-k}u_m\left(\frac{2^m - m}{k}\right) \quad (k \geq 1, k \leq 2^m - m), \quad (54)$$

which is (19), and the finite-defect formula in the form (16). Uniqueness of the intrinsic core prevents double counting, and the bound is uniform over the shape of the core. The two manuscripts add the following remarks to these background statements. Collapse and rigidity are classical facts of the graph representation of finite sets [6, 7]; they explain why counting source-neighborhood families requires no division by automorphism groups (05). The upper

source bound is explicitly on printed page 29 of Tomescu's thesis [7], and the construction and the OEIS formula are background to the question solved here (01). The covering condition $\bigcup(C \cup L) = C$ matters because it has a small failure probability when L is large (01). The source-sensitive enumeration is classical, the OEIS entries contain Johnston's implementation of the sieve and core recurrence [2, 3], and the identities are re-proved so that all normalizations and the exact computations can be checked within the manuscripts, where they also provide exact independent checks of the asymptotic reasoning (both). The covering estimate is useful even though $m2^{-k}$ can exceed one at small parameters: the upper error inequality is still true, while its asymptotic use has k exponentially larger than m (05). The analytic work below uses the positive approximation (54) instead of the cancellation in (16), whose alternating terms are an exact identity, not an asymptotic series (05, 01).

Remark 15.1 (Routes that differ from Part I's). (i) Manuscript 01's collapse proof assigns $x_v = \{x_w : v \rightarrow w\}$ and argues by set-theoretic rank: assignments at lower ranks are distinct by induction, extensionality separates those at the next rank, and different ranks give different sets. (ii) Its support construction takes *any* transitive core C of size m and any k -family of available subsets containing C itself (available because $C \notin C$); every core element is then covered. Part I and manuscript 05 use a directed path as the core. (iii) Manuscript 05 proves the inversion (14) directly: if $f_t = \sum_{s \geq t} \binom{s}{t} u(n, s)$, the coefficient of $u(n, j)$ in $\sum_{t \geq s} (-1)^{t-s} \binom{t}{s} f_t$ is $\binom{j}{s} \sum_{t=s}^j (-1)^{t-s} \binom{j-s}{t-s}$, which vanishes unless $j = s$; and it obtains the total recurrence (15) from $\sum_{t=1}^s (-1)^{t-1} \binom{s}{t} = 1$ for $s \geq 1$, since every nonempty acyclic graph has a source. Part I and manuscript 01 use the polynomial $P_n(z) = \sum_s u(n, s) z^s$. (iv) Manuscript 05 bounds the covering failure by $\binom{2^{m-1}-m}{k} / \binom{2^m-m}{k} \leq ((2^{m-1}-m)/(2^m-m))^k \leq 2^{-k}$.

15.1 The chain lower bound for the core count

The core count affects the singular strength even though it does not affect the exponential radius of the boundary sequence (01). The following two-sided bound is the only information about u_m that the analytic arguments use. Its lower half is new to this report.

Lemma 15.2 (Elementary core-count bounds; sources 05 and 01). *For every $m \geq 1$,*

$$2^{\binom{m-1}{2}} \leq u_m \leq 2^{\binom{m}{2}}. \quad (55)$$

Consequently $\log u_m = \frac{h}{2}m^2 + O(m)$.

Proof. The upper bound is Lemma 2.3. For the lower bound, use vertices $0, \dots, m-1$. Require the arcs $i \rightarrow i-1$ for $1 \leq i < m$ and choose each arc $i \rightarrow j$ with $j < i-1$ independently. The largest member of $N^+(i)$ is $i-1$, so the neighborhoods are distinct. The required path makes the reachability order total; hence the topological order is unique. An isomorphism must preserve that order and therefore every vertex. Different arc choices are nonisomorphic, giving $2^{0+1+\dots+(m-2)}$ classes. There is no factorial loss in this construction. Taking logarithms proves the final assertion. $\square$

Second route for the rigidity step (source 01). In the same graphs, the longest directed path starting at vertex i has length exactly i . These lengths are graph invariants and are all different, so an isomorphism between two such graphs preserves every vertex index, and different choices of optional arcs give different classes. There are $\sum_{i=1}^{m-1} (i-1) = (m-1)(m-2)/2$ optional arcs. Vertex 0 alone has the empty neighborhood. $\square$

Remark 15.3 (source 01). The range $m \geq 1$ in (55) matters: $u_0 = 1$, whereas substituting $m = 0$ into the displayed lower power would give 2. All asymptotic block arguments use $m \rightarrow \infty$.

Wagner's work gives much sharper unrestricted enumeration and distributional results [8, 10]. The elementary squeeze above already gives the exact quadratic coefficient needed for the radial laws, with no unspecified enumeration constant; none of the analytic theorems uses a sharp asymptotic formula for u_m .

15.2 A direct proof of the deletion recurrence

Part I quotes the source-deletion recurrence (44) from Tomescu's Corollary 2.1.7 [7] and uses it only in its verification program. Both manuscripts prove it; manuscript 01's appendix gives the argument in full, and manuscript 05's Section 9.1 gives the same construction in brief.

Proposition 15.4 (Source deletion; sources 01 and 05). *For $n \geq 2$ and $1 \leq s \leq n$, with $u(1, 1) = 1$ and zero outside the valid source range,*

$$s u(n, s) = (2^{n-s} - (n-1))u(n-1, s-1) + \sum_{j=0}^{n-s-1} \binom{s+j}{j+1} 2^{n-1-s-j} u(n-1, s+j), \quad (56)$$

which is (44) multiplied by s .

Proof. Mark one source in a graph with s sources and delete it. If no new source is exposed, the remaining graph has $s-1$ sources. The deleted vertex must avoid these sources, and its out-neighborhood must differ from all $n-1$ existing neighborhoods. There are $2^{n-s} - (n-1)$ choices. When such a remaining graph exists, the source-support bound guarantees this quantity is nonnegative.

Otherwise let $j+1 \geq 1$ be the number of newly exposed sources. The remaining graph has $s+j$ sources. Choose exactly $j+1$ of them as neighbors of the vertex being restored; choose any subset of the other $n-1-s-j$ vertices as further neighbors. These neighborhoods are automatically distinct from every old neighborhood, because they contain an old source. This gives the summand in (56). Rigidity makes the number of choices of a marked source on the left exactly s . Thus the recurrence is independent of binomial inversion, while counting the same objects. $\square$

16 Dyadic blocks and their analytic profile

Fix d for the remainder of the analytic argument and suppress its subscript in $c = c_d$, $a = c-1$ and $R = R_d$. On a block

$$K = 2^{q-1} < n \leq 2K, \quad m = q + d, \quad (57)$$

one has $q(n) = q$ and $2^m = cK$. For sufficiently large q , define

$$k_0 = K + 1 - m, \quad C_q = \binom{cK - m}{k_0}, \quad A_q = u_m C_q R^{K+1}. \quad (58)$$

All these quantities are positive. Choose a fixed lower block index $q_0 = q_0(d)$ large enough that $k_0 \geq 1$ and $m \leq K/2$ thereafter. The omitted initial coefficients make a polynomial $E_d(z)$. The normalization $u_m C_q$ is the positive leading count of the block, not necessarily the exact count $b_{K+1}^{(d)}$; for $d = 0$ the two coincide. By the covering estimate (54),

$$b_{K+1+j}^{(d)} = u_m \binom{cK - m}{k_0 + j} (1 - \eta_{q,j}), \quad 0 \leq \eta_{q,j} \leq m 2^{-(k_0+j)}. \quad (59)$$

Set

$$P_q(z) = \sum_{j=0}^{K-1} \frac{b_{K+1+j}^{(d)}}{u_m C_q} z^j. \quad (60)$$

The exact block decomposition is

$$B_d(z) = E_d(z) + \sum_{q \geq q_0} A_q(z/R)^{K+1} P_q(z), \quad (61)$$

initially as a formal series and later as a convergent analytic identity. (Manuscript 01 writes the same decomposition in the variable $w = z/R$, with its $P_q(w)$ equal to $P_q(Rw)$ here and its amplitude c_q equal to A_q .)

16.1 The uniform geometric profile

The relevant geometric parameter is safely inside the unit disk:

$$aR = \left(1 - \frac{1}{c}\right)^c < e^{-1} < 1. \quad (62)$$

Thus one may choose a real T with $R < T < 1/a$.

Lemma 16.1 (Quantitative block profile; source 05, source 01's Theorem 5.1). *For any fixed $0 < T < 1/a$, uniformly on $|z| \leq T$,*

$$P_q(z) = g(z) + O_{d,T}(m/K), \quad g(z) = \frac{1}{1 - az}. \quad (63)$$

Proof. First replace the count in (60) by its binomial proxy. Write

$$v_{q,j} = \frac{\binom{cK-m}{k_0+j}}{\binom{cK-m}{k_0}}, \quad v_{q,0} = 1.$$

For $1 \leq j < K$ the product representation is

$$v_{q,j} = \prod_{s=1}^j r_s, \quad r_s = \frac{aK - s}{K + 1 - m + s}. \quad (64)$$

The ratios are nonnegative and decrease with s . Since $m \leq K/2$,

$$r_s \leq a + O_d(m/K), \quad |r_s - a| \leq C_d \frac{m+s}{K}. \quad (65)$$

Choose $\epsilon > 0$ such that $(a + \epsilon)T < 1$. For sufficiently large q , every r_s is at most $a + \epsilon$. Telescoping the two products gives, for $j \geq 1$,

$$|v_{q,j} - a^j| \leq \frac{C_d(mj + j^2)}{K} (a + \epsilon)^{j-1}. \quad (66)$$

After multiplication by T^j this is summable uniformly in q . The resulting error is $O_{d,T}(m/K)$. The missing geometric tail from $j = K$ onward is at most $(aT)^K/(1 - aT)$ and satisfies the same bound.

By the covering estimate, the error from replacing the actual coefficient by its proxy is at most $m2^{-k_0-j}v_{q,j}$. Its sum on $|z| \leq T$ is bounded by

$$m2^{-k_0} \sum_{j=0}^{K-1} ((a + \epsilon)T/2)^j = O_{d,T}(m2^{-k_0}).$$

This is smaller than m/K for large q . Combining the three bounds proves (63). $\square$

Manuscript 01 states the case $T = R$ in the variable $w = z/R$: with $aR = \rho_d$ in its notation,

$$\sup_{|w| \leq 1} \left| P_q(Rw) - \frac{1}{1 - aRw} \right| = O_d\left(\frac{m+1}{K}\right) = O_d(q2^{-q}), \quad (67)$$

so the convergence includes the whole circle of convergence and is stronger than convergence of any fixed number of coefficients. Its proof is the same telescoping argument, in the normalized coefficients $v_{q,j}R^j$, whose consecutive ratios it writes exactly: for $0 \leq \ell \leq K-2$,

$$Rr_{\ell+1} = R \frac{aK - 1 - \ell}{K + 2 + \ell - m}, \quad Rr_{\ell+1} - aR = R \frac{am - (2a+1) - c\ell}{K + 2 + \ell - m}, \quad (68)$$

so that $|Rr_{\ell+1} - aR| \leq C_d(m + \ell + 1)/K$ once $K \geq 2m$, with a constant C_d ; these ratios decrease with ℓ , start near $aR < 1$, and are eventually bounded by a fixed $\rho_0 \in (aR, 1)$. The covering error contributes at most $\delta_q/(1 - \rho_0)$ with $\delta_q = m2^{-k_0}$, and the missing tail at most $(aR)^K/(1 - aR)$.

The coefficients of B_d are eventually all positive. The lacunary structure is in the *locations of dominant blocks*. Lemma 16.1 makes that concentration usable: it replaces each long block by a common nonvanishing analytic profile with a uniform error.

16.2 The size of a block amplitude

Lemma 16.2 (Quadratic logarithmic amplitudes; source 05). *There is a constant $C_d > 0$ such that, for every sufficiently large q ,*

$$\exp(hq^2/2 - C_dq) \leq A_q \leq \exp(hq^2/2 + C_dq). \quad (69)$$

In particular, $A_q \rightarrow \infty$ and $\log A_q = o(K)$.

Proof. Falling factorial cancellation gives the exact identity

$$\frac{C_q}{\binom{cK}{K}} = \frac{aK (K)_{\underline{m-1}}}{(cK)_{\underline{m}}} = ac^{-m} \frac{\prod_{i=0}^{m-2} (1 - i/K)}{\prod_{i=0}^{m-1} (1 - i/(cK))}. \quad (70)$$

The logarithm of the product quotient is $O_d(m^2/K)$, so this quotient lies between $1/2$ and 2 for large q .

The binomial probability distribution with parameters cK and $1/c$ has a mode at K . Its mass there is $\binom{cK}{K} R^K$. The largest of its $cK + 1$ masses is between $1/(cK + 1)$ and 1 , hence

$$\frac{1}{cK + 1} \leq \binom{cK}{K} R^K \leq 1. \quad (71)$$

Combining (70)–(71) gives

$$\frac{Ra}{2c^m(cK + 1)} u_m \leq A_q \leq \frac{2Ra}{c^m} u_m. \quad (72)$$

Now apply Lemma 15.2, $m = q + d$, and $\log K = (q - 1)h$. Every contribution other than $hq^2/2$ is $O_d(q)$. $\square$

The argument does not require Stirling's formula. When a more precise normalization is wanted, Stirling's expansion gives the following, which both manuscripts state (manuscript 01 as its Proposition 5.2, with the proof below; manuscript 05 as a display).

Proposition 16.3 (Stirling form of the amplitude; sources 05 and 01). *For fixed d , as $q \rightarrow \infty$,*

$$A_q = u_m c^{-m} R \sqrt{\frac{ca}{2\pi K}} \left(1 + O_d \left(\frac{m^2}{K} \right) \right). \quad (73)$$

Consequently

$$\log A_q = \frac{h}{2} q^2 + O_d(q), \quad A_q \rightarrow \infty. \quad (74)$$

Proof (source 01). In (70), since $m = O_d(\log K)$, the product estimates $(K)_{m-1} = K^{m-1}(1 + O(m^2/K))$ and $(cK)_{\underline{m}} = (cK)^m(1 + O_d(m^2/K))$ follow by summing $\log(1 - j/K)$ and its counterpart. Stirling's formula, with its usual real positive remainder [12], gives

$$\left(\frac{cK}{K} \right) = R^{-K} \sqrt{\frac{c}{2\pi aK}} \left(1 + O_d(K^{-1}) \right). \quad (75)$$

Multiplying (70) and (75) by $u_m R^{K+1}$ yields (73). Taking logarithms and using Lemma 15.2 gives $\log A_q = \frac{h}{2} m^2 - m \log c - \frac{1}{2} \log K + O_d(m) = \frac{h}{2} q^2 + O_d(q)$; the positive quadratic term proves divergence. $\square$

Section 23 gives a first correction to (73).

17 Positive dyadic sums and concentration of block mass

The following purely analytic estimates will be applied to the amplitudes A_q . They also explain why a small uniform block error can be summed near the boundary with a quantitative remainder.

Lemma 17.1 (A discrete Laplace estimate; source 05, source 01's Lemma 7.1). *Suppose $A_q > 0$ for $q \geq q_0 \geq 1$ and*

$$e^{hq^2/2-Cq} \leq A_q \leq e^{hq^2/2+Cq}, \quad h = \log 2, \quad (76)$$

after enlarging C to include finitely many initial indices. Put

$$K_q = 2^{q-1}, \quad S(t) = \sum_{q \geq q_0} A_q e^{-t(K_q+1)}, \quad L = \log(1/t).$$

Then $S(t)$ is finite for $t > 0$, and

$$\log S(t) = \frac{L^2}{2h} + \frac{L \log L}{h} + O_C(L) \quad (t \downarrow 0). \quad (77)$$

Proof. Finiteness follows because $t2^{q-1}$ eventually dominates any quadratic in q . The harmless common factor e^{-t} may be removed in the estimates. Let $X = L + \log L$ and $q_* = \lfloor X/h \rfloor + 1$. Then

$$\frac{L}{2} < tK_{q_*} \leq L, \quad q_* = X/h + O(1). \quad (78)$$

The single q_* term yields

$$\log S(t) \geq \frac{h}{2} q_*^2 - Cq_* - L - t = \frac{X^2}{2h} - O_C(L).$$

For the upper bound set $Q = \lceil X/h \rceil + 4$, so $tK_Q \geq 8L$. For all sufficiently large L , $hQ + C \leq 2L$ and $h \leq L$. Put $M = \exp(hQ^2/2 + CQ)$. The terms with $q \leq Q$ have sum at most QM . For $q = Q + j$, $j \geq 0$, their upper envelope divided by M has logarithm at most

$$2Lj + \frac{h}{2}j^2 - 8L2^j \leq -5L2^j.$$

Here $j \leq 2^j$ and $j^2 \leq 2^{j+1}$ were used. This tail is summable and at most M after increasing L if needed. Therefore $S(t) \leq (Q+1)M$, and $\log S(t) \leq X^2/(2h) + O_C(L)$. Finally $(\log L)^2 = O(L)$, so expanding X^2 proves the statement. $\square$

Second proof of the upper bound (source 01). Put $x = (L + \log L)/h$, $Q = \lceil x \rceil + 3$ and $f(q) = \frac{h}{2}q^2 + Cq - t2^{q-1}$. The head $q \leq Q$ has $O(Q)$ terms, each at most $\exp(\frac{h}{2}Q^2 + CQ)$, whose logarithm is $\mathcal{T}(L) + O(L)$, the term $(\log L)^2$ being absorbed. For the tail, $f(q+1) - f(q) = hq + \frac{h}{2} + C - t2^{q-1}$. At $q = Q$ the negative exponential term has magnitude at least $4L$, whereas $hQ = L + \log L + O(1)$; thus the difference is at most $-L$ for all sufficiently large L . Its subsequent differences are $h - t2^{q-1} < 0$, so the same bound holds for every $q \geq Q$, and the tail is at most $e^{f(Q)}/(1 - e^{-L})$, which has the same logarithmic upper bound as the head. For the lower bound manuscript 01 takes the single term $q = \lfloor x \rfloor$, for which $\frac{h}{2}q^2 = \mathcal{T}(L) + O(L)$, $t2^{q-1} \leq L/2$ and $q = O(L)$. $\square$

Lemma 17.2 (Negligible early blocks and a weighted error bound; source 05). *Under the hypotheses of Lemma 17.1, let q_* be as in (78). For any prescribed $A > 0$, one can choose a fixed integer $D = D(C, A)$ such that*

$$\frac{\sum_{q_0 \leq q \leq q_* - D} A_q e^{-t(K_q+1)}}{S(t)} = O_{C,A}(t^A). \quad (79)$$

For every fixed $p > 0$,

$$\frac{1}{S(t)} \sum_{q \geq q_0} \left(\frac{q}{K_q} \right)^p A_q e^{-t(K_q+1)} = O_{C,p}(t^p). \quad (80)$$

Proof. Use the q_* term as a lower bound for the denominator. Monotonicity of the upper envelope bounds the numerator of (79) by $q_* \exp(h(q_* - D)^2/2 + C(q_* - D))$. Thus the logarithm of its ratio to that lower bound is at most

$$\begin{aligned} \log q_* - hDq_* + \frac{h}{2}D^2 + 2Cq_* - CD + L + t \\ = \left(-D + \frac{2C}{h} + 1 \right) L + O_{C,D}(\log L). \end{aligned}$$

Taking $D > A + 2C/h + 2$ proves (79).

For (80), choose such a D with $A > p$. On $q \geq q_* - D$, the function $q/2^{q-1}$ is eventually decreasing, and (78) gives

$$\frac{q}{K_q} \leq \frac{q_* - D}{K_{q_* - D}} = O_D(t).$$

This portion contributes $O(t^p)$. On earlier indices q/K_q is bounded by an absolute constant, while their relative mass is $O(t^A)$. This proves the weighted estimate. $\square$

The dominant block indices are therefore near $(L + \log L)/\log 2$. Manuscript 05 uses only the one-sided concentration proved above; no unproved smooth approximation to the discrete maximizing index is needed. (Section 21.2 sharpens this: along most radii a single block carries almost all of the mass.)

18 Natural boundaries and radial asymptotics

18.1 Convergence inside the stated circle

Choose $R < T < 1/a$. Lemma 16.1 makes P_q uniformly bounded on $|z| \leq T$, and Lemma 16.2 bounds A_q by a quadratic exponential in q . For $|z| \leq rR$ with $r < 1$, the series in (61) is dominated by

$$C \sum_{q \geq q_0} \exp(hq^2/2 + C_d q) r^{2^{q-1}+1},$$

which converges. Thus it defines an analytic function throughout $|z| < R$ and agrees there with the original generating series.

Let $S_d(t)$ be the sum in Lemma 17.1 formed from these A_q . Lemma 17.2, with $p = 1$ and $m = q + d = O_d(q)$, gives

$$\sum_{q \geq q_0} \frac{m}{K} A_q e^{-t(K+1)} = O_d(t S_d(t)). \quad (81)$$

The polynomial E_d is bounded near $|z| = R$, whereas (77) shows $S_d(t)$ grows faster than every power of $1/t$. In particular $E_d/S_d = O(t)$.

Substituting the uniform profile in (61) on the positive radius yields

$$B_d(Re^{-t}) = g(Re^{-t}) S_d(t) + O_d(t S_d(t)). \quad (82)$$

Since $g(R) > 0$, this tends to infinity. The radius of convergence is at most R , and hence is exactly R . Taking logarithms and using Lemma 17.1 proves (50).

Second route to the radius (source 01). At the block starts, $b_{K+1}^{(d)} R^{K+1} = A_q(1 - \eta_{q,0})$ by (59), and $\log(A_q(1 - \eta_{q,0}))/K \rightarrow 0$ by (74). Therefore $(b_{K+1}^{(d)})^{1/(K+1)} \rightarrow R^{-1}$, so the radius is exactly R ; this recovers (34) without Part I's full entropy theorem. Convergence for $|z| < R$ follows as above from $\sum_q A_q r^{2^{q-1}+1} \leq \sum_q \exp(C_d q^2 - |\log r| 2^{q-1}) < \infty$ for $0 < r < 1$. $\square$

18.2 Alignment at dyadic roots

Fix ζ with $\zeta^{2^j} = 1$. For every $q \geq j + 1$, $K = 2^{q-1}$ is divisible by its order, so

$$(e^{-t} \zeta)^{K+1} = \zeta e^{-t(K+1)}. \quad (83)$$

Only finitely many blocks fail this relation; their contribution is a polynomial bounded as $t \downarrow 0$. The same uniform block error and (81) therefore give

$$B_d(Re^{-t} \zeta) = \zeta g(Re^{-t} \zeta) S_d(t) + O_{d,\zeta}(t S_d(t)). \quad (84)$$

Divide by (82). The denominators stay bounded away from zero, and replacing e^{-t} by 1 changes g by $O_d(t)$. This proves (48). It also proves (50) for the complex radial magnitude and then (51), because $L^2/(2h)$ dominates ML .

18.3 Why dense radial divergence gives a natural boundary

The nonzero limit in (48) and the positive radial divergence show that B_d is unbounded on the inward radius ending at $R\zeta$, for every dyadic root ζ . It cannot be holomorphically continued through such a point: a holomorphic extension would be bounded on a smaller closed neighborhood of that point.

Dyadic roots of unity are dense on the unit circle. If the original analytic germ extended through any point of $|z| = R$, it would extend through a small arc containing a dyadic point, contradicting the preceding paragraph. This proves Theorem 14.2.

The stronger growth (51) also excludes meromorphic continuation at each dyadic point. A meromorphic extension has either a removable singularity or a pole of finite order locally; neither can have this radial growth. Density then rules out meromorphic continuation through any arc. (Manuscript 01 argues the last step differently: a smaller arc would contain infinitely many dyadic blow-up points, incompatible with the locally discrete poles of a meromorphic function.) These boundary points are not isolated singularities, so the terminology “isolated essential singularity” does not apply.

18.4 A general transfer principle (source 01)

Manuscript 01 isolates the analytic step from the graph enumeration. The lemma gives the limit (49) without a rate, and applies to any series built from coherent dyadic blocks.

Lemma 18.1 (Coherent dyadic blocks; source 01). *Suppose, for $|w| < 1$ and an integer $q_0 \geq 1$,*

$$F(w) = E(w) + \sum_{q \geq q_0} \alpha_q w^{2^{q-1}+1} \Pi_q(w),$$

where E and every Π_q are polynomials, $\alpha_q > 0$, and the series converges locally uniformly. Suppose $\Pi_q \rightarrow \Pi$ uniformly on $|w| \leq 1$, $\Pi(1) > 0$, and

$$S(r) = \sum_{q \geq q_0} \alpha_q r^{2^{q-1}+1} \rightarrow \infty \quad (r \uparrow 1).$$

Then for every dyadic root of unity ζ ,

$$\frac{F(r\zeta)}{S(r)} \rightarrow \zeta \Pi(\zeta), \quad \frac{F(r)}{S(r)} \rightarrow \Pi(1). \quad (85)$$

If $\Pi(\zeta) \neq 0$ for every dyadic root of unity ζ , then the unit circle is a natural boundary for F .

Proof. Fix a dyadic root ζ . For all sufficiently large q , $\zeta^{2^{q-1}} = 1$. After moving finitely many blocks into the bounded head, the sum at $r\zeta$ is

$$\zeta \sum_q \alpha_q r^{2^{q-1}+1} \Pi_q(r\zeta).$$

Given $\delta > 0$, choose Q such that $\sup_{|w| \leq 1} |\Pi_q(w) - \Pi(w)| < \delta$ for $q \geq Q$. After division by $S(r)$, the contribution of the finitely many indices below Q tends to zero. The error from the remaining indices is at most δ , because their normalized positive weights sum to at most one. Finally $\Pi(r\zeta) \rightarrow \Pi(\zeta)$. This proves the first limit; the case $\zeta = 1$ proves the second.

If $\zeta \Pi(\zeta) \neq 0$, the first limit implies $|F(r\zeta)| \rightarrow \infty$. A holomorphic continuation through ζ would be bounded on a smaller neighborhood and is therefore impossible. Every open arc contains such a dyadic point. A continuation through any point of the unit circle would contain an arc and would continue through one of those points, a contradiction. $\square$

Second proof of Theorem 14.2, without the rate (source 01). By (74), $A_q \rightarrow \infty$; in particular $\sum_q A_q = \infty$, and monotonicity of the nonnegative terms gives $S(r) \rightarrow \infty$. Apply Lemma 18.1 to $F(w) = B_d(Rw)$ with $\alpha_q = A_q$ and $\Pi_q(w) = P_q(Rw)$. By (67) the limiting profile is $\Pi(w) = 1/(1 - aRw)$, which is nonzero on the closed unit disk. Hence

$$\frac{B_d(Rr\zeta)}{S(r)} \rightarrow \frac{\zeta}{1 - aR\zeta}, \quad \frac{B_d(Rr)}{S(r)} \rightarrow \frac{1}{1 - aR},$$

whose quotient is (49), and the circle $|z| = R$ is a natural boundary. With Lemma 17.1 and (74), $B_d(Re^{-t}) \sim S_d(t)/(1-aR)$ gives $\log B_d(Re^{-t}) = \mathcal{T}(L) + O_d(L)$; at a fixed dyadic ζ the quotient in (49) tends to a nonzero constant, so its logarithmic magnitude is bounded and convergent, which gives Theorem 14.3. Finally $\log(t^M |B_d(Re^{-t}\zeta)|) = \mathcal{T}(L) - ML + O_{d,\zeta}(L) \rightarrow +\infty$. $\square$

Example 18.2 (The first two defects; sources 05 and 01). For $d = 0$ one has $c = 2$, $a = 1$, and $R = 1/4$. Therefore

$$\frac{B_0(\frac{1}{4}e^{-t}\zeta)}{B_0(\frac{1}{4}e^{-t})} = \frac{3\zeta}{4-\zeta} + O_\zeta(t). \quad (86)$$

For $\zeta = -1$ the limit is $-3/5$; for $\zeta = i$ it is $(-3 + 12i)/17$. The negative real direction exhibits cancellation between individual coefficients, but that cancellation has an explicit nonzero limiting effect; the common phase of late dyadic blocks is what makes the limit determinable (manuscript 01). For $d = 1$, $c = 4$, $a = 3$, and $R = 27/256$. The radius becomes smaller, while the first two terms of the logarithmic radial growth remain exactly the same.

Remark 18.3 (What the boundary theorem says; source 01). The conclusion prohibits analytic continuation through every point of the circle. The explicit radial blow-up has been proved at the dense set of points $R\zeta$, ζ a dyadic root of unity. It does not assert pointwise radial blow-up at every irrational angle. Also, these are nonisolated boundary singularities; the classification of isolated singularities into removable points, poles and essential singularities is not directly applicable.

Remark 18.4 (Why the error stops at $O(L)$; sources 01 and 05). The dominant block indices lie around

$$q \approx \frac{L + \log L}{\log 2}. \quad (87)$$

Changing q by one changes 2^{q-1} by a factor of two, and the core bounds (55) allow a linear uncertainty in $\log u_m$, which contributes at order L . Thus (50) does not imply an expansion with a fixed next coefficient times L . Both manuscripts add that the competition between adjacent blocks “remains visible at order L ”, so that a smooth saddle alone cannot justify a constant coefficient at that order; they argue this heuristically. It is stated as a question in Section 24, where Proposition 24.1 [write] proves it under a hypothesis on u_m and gives unconditional bounds for the coefficient.

19 Large boundary values on every arc

The radial theorem already implies singularities at a dense set. We now prove the stronger comparison in Theorem 14.4. The issue is quantitative: one very large value on a tiny interval would not by itself give a fixed proportion of the radial L^p size. Manuscript 05 uses a growing grid of dyadic points and a derivative bound.

19.1 A derivative bound from positivity

Write $F_t = B_d(Re^{-t}) > 0$ and

$$D_t = \sum_{n \geq 1} n b_n^{(d)} R^n e^{-tn}. \quad (88)$$

Absolute convergence inside the disk implies

$$\left| \frac{\partial}{\partial \theta} B_d(Re^{-t+i\theta}) \right| \leq D_t.$$

Let $f(t) = \log F_t$. Differentiation of the positive series gives $-f'(t) = D_t/F_t$ and $f''(t) \geq 0$: the latter is the variance of the index under the probabilities proportional to $b_n^{(d)} R^n e^{-tn}$. Convexity implies

$$f(t/2) - f(t) \geq \frac{t}{2} \frac{D_t}{F_t}.$$

By (50), with L replaced by $L + h$ at $t/2$, $f(t/2) - f(t) = O_d(L)$. Consequently

$$D_t \leq C_d \frac{L}{t} F_t \quad (89)$$

for sufficiently small t .

19.2 A moving dyadic grid

Let $q_* = \lfloor (L + \log L)/h \rfloor + 1$ as before. Choose a fixed large D so that the relative weight of all blocks $q < q_b = q_* - D$ is $O_d(t)$, using Lemma 17.2. Set $K_b = 2^{q_b-1}$. Then

$$2^{-D-1} \frac{L}{t} < K_b \leq 2^{-D} \frac{L}{t}. \quad (90)$$

Consider all K_b roots of unity ζ with $\zeta^{K_b} = 1$, and their periodic angular lifts $\theta_j = 2\pi j/K_b$ for $j \in \mathbb{Z}$. For every block $q \geq q_b$, its K_q is a multiple of K_b , and the alignment (83) holds simultaneously for all these ζ .

The early blocks have modulus at most a fixed constant times their positive amplitude sum, because the P_q are uniformly bounded. Their contribution divided by $S_d(t)$ is $O_d(t)$; the finite polynomial is smaller still. Also $\sup_{q \geq q_b} (m/K_q) = O_d(t)$ by (90). The profile estimate is uniform in ζ , so

$$\sup_{\zeta^{K_b}=1} \left| \frac{B_d(\text{Re}^{-t}\zeta)}{S_d(t)} - \zeta g(\text{Re}^{-t}\zeta) \right| = O_d(t). \quad (91)$$

The rational profile has modulus at least $1/(1 + aR)$ on the closed disk $|z| \leq R$. Combining this with (82), there is $\kappa_d > 0$ such that

$$|B_d(\text{Re}^{-t}\zeta)| \geq \kappa_d F_t \quad \text{for every } \zeta^{K_b} = 1 \quad (92)$$

when t is sufficiently small.

This estimate is uniform over a grid whose order grows with t . It is proved directly by discarding negligible early blocks; it does not assume that the constant in the fixed-root Theorem 14.2 was uniform in the order of the root.

19.3 Proof of the local comparison

By (89), the modulus stays at least $\kappa_d F_t/2$ within angular distance

$$\delta_t = \eta_d \frac{t}{L} \quad (93)$$

of each grid point, for a fixed sufficiently small $\eta_d > 0$. Choose η_d also small enough that these intervals are disjoint; this is possible by (90), since the grid spacing is $2\pi/K_b$.

Every fixed open interval I contains a closed interval J of positive length in its interior. The number of grid points in J is at least $|J|K_b/(4\pi)$ for large K_b . Their intervals of radius δ_t lie in I for small t . Their total length is bounded below by a positive constant depending only on d and I , because $K_b \delta_t$ is bounded below in (90)–(93). Hence

$$\int_I |B_d(\text{Re}^{-t+i\theta})|^p d\theta \geq c_{d,I} (\kappa_d F_t/2)^p.$$

For the upper bound, coefficient positivity gives $|B_d(Re^{-t+i\theta})| \leq F_t$, so the integral is at most $|I|F_t^p$. This proves (52) for every $p > 0$; no norm inequality restricted to $p \geq 1$ has been used. Taking logarithms and using (50) proves (53).

Second proof of the logarithmic law (53) (source 01). Set $\tilde{B}(w) = B_d(Rw) = \sum_{n \geq 1} \beta_n w^n$, where $\beta_n = b_n^{(d)} R^n \geq 0$, and let $r = e^{-t}$, $s = e^{-t/2}$. Choose an angle θ_0 in the interior of I with $\zeta = e^{i\theta_0}$ a dyadic root of unity. Equation (49) supplies a constant $\kappa_\zeta > 0$ such that, for all sufficiently small t ,

$$|\tilde{B}(r\zeta)| \geq \kappa_\zeta \tilde{B}(r). \quad (94)$$

Nonnegativity and absolute convergence allow differentiation inside the disk. Since $xe^{-tx/2} \leq 2/(et)$ for $x \geq 0$,

$$\sup_{\theta} \left| \frac{\partial}{\partial \theta} \tilde{B}(re^{i\theta}) \right| \leq \sum_{n \geq 1} n \beta_n r^n \leq \frac{2}{et} \sum_{n \geq 1} \beta_n s^n = \frac{2\tilde{B}(s)}{et}. \quad (95)$$

Take $\delta = \kappa_\zeta et \tilde{B}(r)/(8\tilde{B}(s))$. Because $\tilde{B}(r) \leq \tilde{B}(s)$, this width tends to zero and eventually $(\theta_0 - \delta, \theta_0 + \delta) \subset I$. On that interval, (95) changes $\tilde{B}$ by at most $\kappa_\zeta \tilde{B}(r)/4$; in particular $|\tilde{B}(re^{i\theta})| \geq \kappa_\zeta \tilde{B}(r)/2$. For every fixed $p > 0$, integration over this interval gives

$$\int_I |\tilde{B}(re^{i\theta})|^p d\theta \geq C_{\zeta,p} t \frac{\tilde{B}(r)^{p+1}}{\tilde{B}(s)}, \quad (96)$$

while the upper bound is simply

$$\int_I |\tilde{B}(re^{i\theta})|^p d\theta \leq |I| \tilde{B}(r)^p. \quad (97)$$

The radial growth theorem gives $\log \tilde{B}(r) = \mathcal{T}(L) + O_d(L)$ and $\log \tilde{B}(s) = \mathcal{T}(L + \log 2) + O_d(L)$, and direct expansion shows $\mathcal{T}(L + \log 2) - \mathcal{T}(L) = O(L)$. Taking logarithms in (96) and (97) places the integral between $\exp(p\mathcal{T}(L) - O_{d,I,p}(L))$ and $\exp(p\mathcal{T}(L) + O_{d,I,p}(L))$, as required. This route uses a single dyadic point; it gives the logarithmic law but not the two-sided comparison (52), whose lower bound loses the factor $t\tilde{B}(r)/\tilde{B}(s)$. $\square$

Remark 19.1 (Strong natural boundary; source 05). Breuer and Simon call the unit circle a strong natural boundary when the radial local L^1 integrals are not uniformly bounded on any nonempty open arc [11]. Apply this terminology to $w \mapsto B_d(R_d w)$. Theorem 14.4 proves that condition, and more: there is an actual limit $+\infty$, a comparison with the positive radial value, and a quantitative logarithmic law for every positive exponent p .

Remark 19.2 (source 01). The theorem excludes a bounded radial L^p integral on every fixed nonempty boundary arc, and in particular excludes membership in every Hardy class H^p , $p > 0$, after normalization to the unit disk. For $0 < p < 1$, only the integral, not a norm inequality, is used. The conclusion makes no assertion about the existence or values of radial limits at almost every individual angle.

20 Consequences for generating functions and defect marking

20.1 Algebraicity, differential equations, and labeling

Corollary 20.1 (sources 05 and 01). *For each fixed d , B_d is neither algebraic nor D -finite over $\mathbb{C}(z)$. The sequences $b_n^{(d)}$ and $n!b_n^{(d)}$ are not P -recursive. The ordinary generating function of $n!b_n^{(d)}$ has radius zero.*

Proof. An algebraic function has only finitely many possible finite branch points and poles: away from the zeros of the leading coefficient and discriminant of a square-free defining polynomial, the implicit function theorem supplies continuation. A solution of a linear differential equation with polynomial coefficients continues locally at every point where the leading coefficient is nonzero, by the existence theorem for analytic linear differential equations. Neither can have the natural boundary just proved. $\square$

A polynomial-coefficient recurrence for the coefficients translates, by multiplying by powers of z and summing, to a linear differential equation with polynomial coefficients for the generating function; initial terms contribute only a polynomial. Thus $b_n^{(d)}$ is not P-recursive. If $n!b_n^{(d)}$ had such a recurrence, write it with forward shifts. Division by $n!$ would replace each factorial quotient by a polynomial in n and give one for $b_n^{(d)}$. Finally the finite positive radius R_d implies a subsequence with $(b_n^{(d)})^{1/n} \rightarrow R_d^{-1} > 0$, while $(n!)^{1/n} \rightarrow \infty$. The labeled ordinary series therefore has radius zero. $\square$

Second route for non-algebraicity (source 01). An algebraic function and all its derivatives belong to one finite-dimensional algebraic extension of $\mathbb{C}(z)$. Linear dependence of sufficiently many derivatives gives a differential equation over $\mathbb{C}(z)$, and clearing denominators gives a polynomial-coefficient equation; so an algebraic series is D-finite, and the D-finite case applies. Manuscript 01 also notes that the equivalence between D-finite series and P-recursive coefficient sequences follows by coefficient extraction in the differential equation and the converse construction. $\square$

The non-P-recursive conclusion was already proved in Part I (Theorem 8.2, through the exponential-jump Lemma 8.1); the natural boundary strengthens its complex-analytic explanation. It does not exclude nonlinear algebraic differential equations. That is a separate question (Section 24).

For labeled counts $\ell_n^{(d)} = n!b_n^{(d)}$, the exponential generating function is exactly

$$\sum_{n \geq 1} \ell_n^{(d)} \frac{z^n}{n!} = B_d(z).$$

Thus the same boundary theorem describes labeled enumeration in its natural exponential normalization; the ordinary generating function of $\ell_n^{(d)}$, of radius zero, must not be confused with B_d (manuscript 01).

20.2 Finite combinations and elementary transformations

Corollary 20.2 (sources 05 and 01). *The radii are strictly decreasing in d : $R_0 > R_1 > \dots$. If $\gamma_D \neq 0$, the finite linear combination $\sum_{d=0}^D \gamma_d B_d(z)$, with complex coefficients, has natural boundary $|z| = R_D$. Adding to B_d any function holomorphic on a larger disk also preserves its natural boundary.*

Proof. As a function of real $c > 1$,

$$\frac{d}{dc} \log \frac{(c-1)^{c-1}}{c^c} = \log(1 - 1/c) < 0,$$

which gives the monotonicity. All terms with $d < D$ are analytic on a larger disk than $|z| < R_D$, so continuation of the combination would give continuation of the nonzero multiple $\gamma_D B_D$, which is impossible. This argument permits complex coefficients because only one summand has the smallest radius. The perturbation statement is the same subtraction argument. $\square$

Changing finitely many coefficients does not alter a natural boundary. Neither does taking a derivative or a primitive: local continuation of a derivative can be integrated on a small disk and matched to the original function by an additive constant, while continuation of a primitive implies that of its derivative. These facts transfer the result to shifted or polynomially weighted enumerations without repeating the block proof (manuscript 05).

20.3 There is no analytic size-and-defect germ

For each n , only finitely many defects are admissible. Thus

$$\mathcal{B}(z, v) = \sum_{n \geq 1} \sum_{d \geq 0} b_n^{(d)} z^n v^d \quad (98)$$

is a well-defined formal series, with a polynomial in v as each coefficient of z^n .

Theorem 20.3 (Positive-fugacity obstruction; source 05). *For every fixed real $v > 0$, the series*

$$\sum_{n \geq 1} \left(\sum_{d \geq 0} b_n^{(d)} v^d \right) z^n \quad (99)$$

has radius zero. Consequently (98) is not a holomorphic germ at $(0, 0)$ and has no nontrivial bidisk of convergence.

Proof. Fix $D \geq 0$. Nonnegativity gives $\sum_d b_n^{(d)} v^d \geq v^D b_n^{(D)}$. Therefore

$$\limsup_{n \rightarrow \infty} \left(\sum_d b_n^{(d)} v^d \right)^{1/n} \geq R_D^{-1}.$$

But

$$R_D = \frac{1}{c_D} \left(1 - \frac{1}{c_D} \right)^{c_D - 1} \sim \frac{1}{ec_D} \rightarrow 0. \quad (100)$$

Since D is arbitrary, the limsup is infinite. A holomorphic germ in two variables would have a convergent power series on some positive-radius bidisk, contradicting any sufficiently small positive choice of v . $\square$

Keeping a defect fixed and summing over defects have different analytic consequences. More generally, let $w_d \geq 0$ be weights independent of n , with $w_d > 0$ for infinitely many d . Arbitrarily strong decay of w_d cannot restore a positive size radius for $\sum_{n,d} w_d b_n^{(d)} z^n$: comparison with each fixed D for which $w_D > 0$ proves this extension. The argument does not assert zero radius for every nonzero *complex* v , where signed or complex cancellation requires separate analysis.

21 Arithmetic sections

21.1 Filtering one residue class of power-of-two step (source 01)

For a step $r \geq 1$ and a residue $0 \leq r_0 < r$ define

$$B_d^{(r, r_0)}(z) = \sum_{\substack{n \geq 1 \\ n \equiv r_0 \pmod{r}}} b_n^{(d)} z^n, \quad G_d^{(r, r_0)}(z) = \sum_{\ell \geq 0} b_{r\ell + r_0}^{(d)} z^\ell. \quad (101)$$

(Here r is a step, not Part I's deletion number.) The convention $b_0^{(d)} = 0$ gives

$$B_d^{(r,r_0)}(z) = z^{r_0} G_d^{(r,r_0)}(z^r). \quad (102)$$

In this subsection $r = 2^s$ with $s \geq 0$. Let $j_0 \in \{0, \dots, 2^s - 1\}$ be the unique residue with $j_0 \equiv r_0 - 1 \pmod{2^s}$.

Theorem 21.1 (Dyadic subsequences retain a natural boundary; source 01). *Let $r = 2^s$. The radii of $B_d^{(r,r_0)}$ and $G_d^{(r,r_0)}$ are respectively R_d and R_d^r , and their circles of convergence are natural boundaries. For every dyadic root of unity ζ ,*

$$\lim_{t \downarrow 0} \frac{B_d^{(r,r_0)}(R_d e^{-t} \zeta)}{B_d^{(r,r_0)}(R_d e^{-t})} = \zeta^{1+j_0} \frac{1 - (a_d R_d)^r}{1 - (a_d R_d)^r \zeta^r}. \quad (103)$$

The logarithm of the positive radial value $B_d^{(r,r_0)}(R_d e^{-t})$ has the same two-term expansion $\mathcal{T}(L) + O_{d,s}(L)$.

Proof. For all sufficiently large q , $K = 2^{q-1}$ is divisible by r . The allowed offsets $j = n - K - 1$ in the block are then exactly $j \equiv j_0 \pmod{r}$. The proof of Lemma 16.1 (or of (67)), restricted to these offsets, gives the uniform limit

$$P_q^{(r,r_0)}(Rw) \longrightarrow \sum_{\ell \geq 0} (aRw)^{j_0+r\ell} = \frac{(aRw)^{j_0}}{1 - (aRw)^r} \quad (|w| \leq 1), \quad (104)$$

where $P_q^{(r,r_0)}$ keeps the terms of P_q with $j \equiv j_0$. Indeed the sum of the absolute coefficient errors over a subset of offsets is no larger than over all offsets. The limiting profile at $w = 1$ is the positive number $(aR)^{j_0}/(1 - (aR)^r)$.

The same block amplitudes A_q and weights $e^{-t(2^{q-1}+1)}$ occur. Lemma 18.1 therefore gives (103), positive radial divergence, and a natural boundary for $B_d^{(r,r_0)}$. It also gives

$$B_d^{(r,r_0)}(R_d e^{-t}) \sim \frac{(aR)^{j_0}}{1 - (aR)^r} S_d(t),$$

which proves the radial expansion.

Absolute convergence and (102) show that $G_d^{(r,r_0)}$ is analytic for $|z| < R_d^r$. If it extended holomorphically through a point z_0 with $|z_0| = R_d^r$, choose any w_0 with $w_0^r = z_0$. Composing the extension with $w \mapsto w^r$ and multiplying by w^{r_0} would extend $B_d^{(r,r_0)}$ through w_0 , contradicting its natural boundary. This proves the remaining claims. $\square$

Example 21.2 (source 01). For the even and odd subsequences of the extremal count, both series

$$\sum_{\ell \geq 0} b_{2\ell}^{(0)} z^\ell, \quad \sum_{\ell \geq 0} b_{2\ell+1}^{(0)} z^\ell$$

have natural boundary $|z| = 1/16$. For the corresponding filtered series in the original variable, the limiting profiles are $aRw/(1 - (aRw)^2)$ and $1/(1 - (aRw)^2)$, respectively, with $aR = 1/4$.

Corollary 21.3 (source 01). *None of $B_d^{(2^s,r_0)}$ or $G_d^{(2^s,r_0)}$ is D-finite. In particular they are not algebraic over $\mathbb{C}(z)$, and every arithmetic subsequence $(b_{2^s\ell+r_0}^{(d)})_{\ell \geq 0}$ is not P-recursive.*

Proof. As for Corollary 20.1: a D-finite function continues analytically away from the finitely many zeros of its leading coefficient, so a full circle of noncontinuability is impossible, and an algebraic series is D-finite. $\square$

The subsequence statement supplies additional non-P-recursive sequences; the case of B_d itself is Corollary 20.1.

21.2 A single dominant block, and sections of every step [write]

Theorem 21.1 uses that every late block starts at the same residue modulo 2^s . For a step with an odd factor the starting residue $K_q + 1 \bmod r$ cycles with q , and both manuscripts leave such sections open (manuscript 05's Question 5: "root-of-unity filters can introduce cancellation"; manuscript 01's Question 4). The writing step closes the natural-boundary part of that question. The input is a comparison of neighbouring blocks that is sharper than the envelope (69); it holds because the ratio u_{m+1}/u_m is known to within a factor $m + 1$.

For $t > 0$ put $L = \log(1/t)$, $X = (L + \log L)/h$, and

$$\omega_q(t) = A_q e^{-t(K_q+1)}, \quad y_0(L) = \lceil X \rceil - X \in [0, 1), \quad q_t = 1 + \lceil X \rceil, \quad (105)$$

so that $S_d(t) = \sum_{q \geq q_0} \omega_q(t)$ and $tK_{q_t} = L 2^{y_0(L)}$.

Lemma 21.4 (A single dominant block; write). *Fix $d \geq 0$ and $\delta \in (0, 1/2]$, and let $c_\delta = (1 - 2^{-\delta})/4$. For all sufficiently small t with $\delta \leq y_0(L) \leq 1 - \delta$,*

$$\sum_{q \neq q_t} \omega_q(t) \leq e^{-c_\delta L} \omega_{q_t}(t). \quad (106)$$

In particular $S_d(t) = \omega_{q_t}(t)(1 + O(e^{-c_\delta L}))$ along such t .

Proof. Neighbouring amplitudes. Proposition 5.1 with $n = m + 1$ and $t = 1$ gives $\sum_s s u(m + 1, s) = u_m(2^m - m)$; every class counted by u_{m+1} has between 1 and $m + 1$ sources, so

$$\frac{2^m - m}{m + 1} \leq \frac{u_{m+1}}{u_m} \leq 2^m - m, \quad \log \frac{u_{m+1}}{u_m} = hm + O(\log m). \quad (107)$$

By (73), $\log A_q = \log u_{q+d} - (q + d) \log c - \frac{1}{2}(q - 1)h + O_d(1)$, with $\log c = (d + 1)h$. Hence $\log A_{q+1} - \log A_q = hq + O_d(\log q)$, and since $K_{q+1} - K_q = K_q$,

$$\log \omega_{q+1}(t) - \log \omega_q(t) = hq - t2^{q-1} + \epsilon_q, \quad |\epsilon_q| \leq C_d \log q \quad (q \geq q_0). \quad (108)$$

Blocks above q_t . Write $q = q_t + k$ with $k \geq 0$. Then $hq = L + \log L + h(1 + y_0 + k)$ and $t2^{q-1} = L2^{y_0+k}$, so the increment (108) equals $L(1 - 2^{y_0+k}) + \log L + h(1 + y_0 + k) + \epsilon_q$. Here $2^{y_0+k} - 1 \geq 2^k(2^\delta - 1)$, while $\log L + h(2 + k) + C_d \log(2X + 2 + k) \leq 2^{k-1}(2^\delta - 1)L$ for all $k \geq 0$ once L is large. So every increment with $q \geq q_t$ is at most $-(2^\delta - 1)L/2$, and $\sum_{q > q_t} \omega_q \leq 2e^{-(2^\delta - 1)L/2} \omega_{q_t}$ for large L .

Blocks below q_t . For $q \leq q_t - 1$ one has $t2^{q-1} \leq L2^{y_0-1} \leq L2^{-\delta}$. If $hq \geq \frac{1}{2}(1 + 2^{-\delta})L + C_d \log q$, the increment is at least $\frac{1}{2}(1 - 2^{-\delta})L > 0$. Otherwise $q \leq (1 + 2^{-\delta})L/(2h) + O(\log L)$, so $q_t - q \geq (1 - 2^{-\delta})L/(2h) - O(\log L)$ and $t2^{q-1} = L2^{y_0+q-q_t} \leq 1$ for large L ; then the increment is at least $hq - 1 - C_d \log q \geq 1$ once $q \geq q_1$, a constant depending on d . Hence ω_q increases on $q_1 \leq q \leq q_t$. At $q = q_t - 1$, $hq = L + \log L + hy_0 \geq L$, so the last increment is at least $L(1 - 2^{-\delta}) - C_d \log q \geq \frac{1}{2}(1 - 2^{-\delta})L$. Therefore $\sum_{q_1 \leq q < q_t} \omega_q \leq q_t e^{-(1 - 2^{-\delta})L/2} \omega_{q_t}$, with $q_t = O(L)$. The finitely many $q_0 \leq q < q_1$ have $\omega_q \leq A_q = O_d(1)$, whereas $\log \omega_{q_t} \geq \frac{h}{2}q_t^2 - C_d q_t - L2^{y_0} - t \geq L^2/(2h) - O_d(L)$ by (69). Since $2^\delta - 1 \geq 1 - 2^{-\delta}$, the three bounds give (106) for small t . $\square$

Theorem 21.5 (Every arithmetic section has a natural boundary; write). *Fix $d \geq 0$, a step $r \geq 1$ and a residue $0 \leq r_0 < r$. Then $B_d^{(r, r_0)}$ has radius R_d and $G_d^{(r, r_0)}$ has radius R_d^r , and both circles of convergence are natural boundaries. More precisely, let $j_q \in \{0, \dots, r - 1\}$ be the residue of $r_0 - 1 - K_q$ modulo r . For every dyadic root of unity ζ and every $\delta \in (0, 1/2]$, as $t \downarrow 0$ with $\delta \leq y_0(L) \leq 1 - \delta$,*

$$B_d^{(r, r_0)}(R_d e^{-t} \zeta) = \zeta \omega_{q_t}(t) \left[\frac{(a_d R_d \zeta)^{j_{q_t}}}{1 - (a_d R_d \zeta)^r} + o(1) \right], \quad (109)$$

and $|B_d^{(r,r_0)}(R_d e^{-t}\zeta)| \rightarrow \infty$ along such t , with logarithm $\mathcal{T}(L) + O_{d,r}(L)$. Consequently every arithmetic subsequence $(b_{r\ell+r_0}^{(d)})_{\ell \geq 0}$ and $((r\ell + r_0)! b_{r\ell+r_0}^{(d)})_{\ell \geq 0}$ is not P -recursive, and $B_d^{(r,r_0)}$, $G_d^{(r,r_0)}$ are neither D -finite nor algebraic.

Proof. In block q the coefficients with $n = K + 1 + j \equiv r_0 \pmod{r}$ are those with $j \equiv j_q \pmod{r}$. Let $P_q^{(r,r_0)}$ keep exactly these terms of P_q . Since $(az)^{j_q}/(1 - (az)^r) = \sum_{j \equiv j_q} (az)^j$, the three error terms in the proof of Lemma 16.1 (proxy against geometric coefficients, geometric tail beyond K , covering error) only lose nonnegative terms when restricted to $j \equiv j_q$, so for a fixed $T \in (R, 1/a)$

$$\sup_{|z| \leq T} \left| P_q^{(r,r_0)}(z) - \frac{(az)^{j_q}}{1 - (az)^r} \right| = O_{d,T}(m/K),$$

uniformly in q , and the $P_q^{(r,r_0)}$ are uniformly bounded there, by M_d say. The block decomposition (61) restricted to $n \equiv r_0$ reads $B_d^{(r,r_0)}(z) = E^{(r,r_0)}(z) + \sum_q A_q(z/R)^{K+1} P_q^{(r,r_0)}(z)$ with a polynomial head. Let $\zeta^{2^j} = 1$. By (83), for $q > j$ the block q contributes $\zeta \omega_q(t) P_q^{(r,r_0)}(Re^{-t}\zeta)$ at $z = Re^{-t}\zeta$; the finitely many other blocks and the head are $O(1)$. Hence

$$\left| B_d^{(r,r_0)}(Re^{-t}\zeta) - \zeta \omega_{q_t}(t) P_{q_t}^{(r,r_0)}(Re^{-t}\zeta) \right| \leq M_d \sum_{q \neq q_t} \omega_q(t) + O(1).$$

By Lemma 21.4 the right side is $O(e^{-c_s L} \omega_{q_t}(t)) + O(1) = o(\omega_{q_t}(t))$. Moreover $P_{q_t}^{(r,r_0)}(Re^{-t}\zeta) = (aRe^{-t}\zeta)^{j_{q_t}}/(1 - (aRe^{-t}\zeta)^r) + O(m/K)$, and replacing e^{-t} by 1 changes this rational value by $O_r(t)$ uniformly in $j_{q_t} < r$. This proves (109). The bracket has modulus at least $(aR)^{r-1}/(1 + (aR)^r) - o(1) > 0$, and $\omega_{q_t}(t) = S_d(t)(1 + o(1)) \rightarrow \infty$ with $\log S_d(t) = \mathcal{T}(L) + O_d(L)$ by Lemma 17.1. Since $y_0(L)$ runs through $[0, 1)$ infinitely often as $L \rightarrow \infty$ (it is continuous between its jumps, at which it returns from 0 to 1), such $t \downarrow 0$ exist, and $B_d^{(r,r_0)}$ is unbounded on the radius to $R\zeta$. As in Section 18, it has no holomorphic continuation through $R\zeta$, and density of the dyadic roots gives the natural boundary. Its coefficients are nonnegative and dominated by those of B_d , so it is analytic in $|z| < R$ and its radius is exactly R . The statements for $G_d^{(r,r_0)}$ follow from (102) exactly as in the proof of Theorem 21.1. A D -finite or algebraic function cannot have a natural boundary (Corollary 20.1), so $(b_{r\ell+r_0}^{(d)})$ is not P -recursive; a polynomial recurrence for $((r\ell + r_0)! b_{r\ell+r_0}^{(d)})$, written with forward shifts and divided by $(r\ell + r_0)!$, would give one for $(b_{r\ell+r_0}^{(d)})$, because $(r(\ell + i) + r_0)!/(r\ell + r_0)!$ is a polynomial in ℓ . $\square$

Remark 21.6 (What Theorem 21.5 does not say; write). For $r = 2^s$, $j_q = j_0$ for all large q and (109) recovers Theorem 21.1 along the restricted radii. When r has an odd factor $r' > 1$, j_q depends on $2^{q-1} \bmod r'$, which is periodic in q with period the order of 2 modulo r' ; the bracket in (109) then changes with q_t . Whether the radial ratio at a dyadic root then has a limit is decided by Corollary 21.7 below; nothing is proved about the radii with $y_0(L)$ near 0, where two blocks compete, beyond the sign change noted after that corollary, or about non-dyadic angles. The theorem is the writing step's; it has not been externally reviewed. Its only inputs are Lemma 16.1, the Stirling form (73), Part I's marked-source identity and the envelope (69). A numerical illustration for $d = 0$ and $r = 3$ is in Section 22.

Dated note (5 October 2026). Until this date the remark's third sentence ended “and nothing is proved about the existence of a radial ratio limit, about the radii with $y_0(L)$ near 0, where two blocks compete, or about non-dyadic angles”, and the next sentence read “The theorem is the writing step's; it has not been independently reviewed.” The first item understated the theorem, which already decides it (Corollary 21.7). The intake's later check of the theorem is recorded in item 13 of Section 24; it is not an external review.

Corollary 21.7 (Radial ratios of a section at a dyadic root; intake). *Fix $d \geq 0$, a step $r \geq 1$, a residue $0 \leq r_0 < r$ and a dyadic root of unity ζ of order 2^e . Let j_q be as in Theorem 21.5, let J be the set of residues that j_q takes for infinitely many q , and put*

$$\kappa_j(\zeta) = \zeta^{1+j} \frac{1 - (a_d R_d)^r}{1 - (a_d R_d \zeta)^r} \quad (0 \leq j < r),$$

so that $\kappa_j(\zeta)/\kappa_{j'}(\zeta) = \zeta^{j-j'}$. Write $\varrho(t) = B_d^{(r,r_0)}(R_d e^{-t}\zeta)/B_d^{(r,r_0)}(R_d e^{-t})$.

- (i) For every $j \in J$ and $\delta \in (0, 1/2]$, $\varrho(t) \rightarrow \kappa_j(\zeta)$ as $t \downarrow 0$ with $\delta \leq y_0(L) \leq 1 - \delta$ and $j_{q_t} = j$, and such t exist arbitrarily close to 0.
- (ii) $\varrho(t)$ has a limit as $t \downarrow 0$ if and only if ζ^{j_q} is eventually constant, that is, 2^e divides $j - j'$ for all $j, j' \in J$. The limit is then the common value of the $\kappa_j(\zeta)$, $j \in J$, and it is attained along all $t \downarrow 0$, with no restriction on $y_0(L)$.

Proof. Write $a = a_d$, $R = R_d$, $B = B_d^{(r,r_0)}$ and $g_j(w) = w^j/(1-w^r)$. Then $g_j(aR) \geq (aR)^{r-1} > 0$ and $\zeta g_j(aR\zeta) = \kappa_j(\zeta)g_j(aR)$. Since $K_q = 2^{q-1}$ is eventually periodic modulo r , so is j_q ; hence J is not empty and $j_q \in J$ for all large q .

(i) Along $\delta \leq y_0(L) \leq 1 - \delta$, (109) at ζ and at 1 gives $B(Re^{-t}\zeta) = \omega_{q_t}(t)[\zeta g_{j_{q_t}}(aR\zeta) + o(1)]$ and $B(Re^{-t}) = \omega_{q_t}(t)[g_{j_{q_t}}(aR) + o(1)]$, and the second bracket is at least $(aR)^{r-1}/2$ for small t ; so $\varrho(t) = \kappa_{j_{q_t}}(\zeta) + o(1)$. Since $X = (L + \log L)/h$ increases continuously to ∞ with L , every large q is q_t for some t with $y_0(L) = 1/2$ (take $X = q - 3/2$); taking the q with $j_q = j$ gives the required t .

(ii) If $\zeta^j \neq \zeta^{j'}$ for some $j, j' \in J$, part (i) gives the two subsequential limits $\kappa_j(\zeta) \neq \kappa_{j'}(\zeta)$, so there is no limit. Conversely, let κ be the common value of $\kappa_j(\zeta)$, $j \in J$, and fix $\epsilon > 0$. As in the proof of Theorem 21.5, for $q > e$ block q contributes $\zeta \omega_q(t)P_q^{(r,r_0)}(Re^{-t}\zeta)$ at $Re^{-t}\zeta$ and $\omega_q(t)P_q^{(r,r_0)}(Re^{-t})$ at Re^{-t} , and $P_q^{(r,r_0)}(Re^{-t}w) = g_{j_q}(aRw) + O_d(m/K) + O_r(t)$ uniformly in $|w| \leq 1$ and q . Choose $Q > e$ with $j_q \in J$ for $q \geq Q$ and the $O_d(m/K)$ term below ϵ there. Because $\zeta g_{j_q}(aR\zeta) = \kappa g_{j_q}(aR)$ for $q \geq Q$, and the head and the blocks $q < Q$ contribute $O_Q(1)$,

$$|B(Re^{-t}\zeta) - \kappa B(Re^{-t})| \leq (1 + |\kappa|)(\epsilon + O_r(t))S_d(t) + O_Q(1).$$

All coefficients of B and of every $P_q^{(r,r_0)}$ are nonnegative, so if $\epsilon \leq (aR)^{r-1}/4$, then for small t $B(Re^{-t}) \geq \sum_{q \geq Q} \omega_q(t)P_q^{(r,r_0)}(Re^{-t}) \geq \frac{1}{2}(aR)^{r-1}(S_d(t) - O_Q(1))$. As $S_d(t) \rightarrow \infty$, $\limsup_{t \downarrow 0} |\varrho(t) - \kappa| \leq 2(1 + |\kappa|)\epsilon/(aR)^{r-1}$ for every small ϵ , so $\varrho(t) \rightarrow \kappa$. $\square$

The converse in (ii) is the argument of Lemma 18.1, with the positive weights $\omega_q(t)g_{j_q}(aR)$; it uses neither Lemma 21.4 nor a restriction on $y_0(L)$. For $r = 2^s$, $J = \{j_0\}$ and (ii) is (103).

Dated note (5 October 2026). The intake added this corollary after its check of the write (item 13 of Section 24), which observed that Theorem 21.5 already answers the radial-ratio part of Remark 21.6 and of item 5 of Section 24. Examples, with $K_q = 2^{q-1}$ reduced modulo r for $q \geq 2$:

- $r = 3$: $K_q \equiv 2, 1$ alternately, so $J = \{0, 1\}$, $\{1, 2\}$, $\{0, 2\}$ for $r_0 = 0, 1, 2$. For $r_0 = 0, 1$ the ratio has no limit at any dyadic $\zeta \neq 1$; for $r_0 = 2$ it has one at $\zeta = -1$ but not at roots of order 4 or more. For $d = 0$ ($a_0 R_0 = 1/4$), $r_0 = 0$ and $\zeta = -1$ it jumps between $\kappa_0(-1) = -63/65$ and $\kappa_1(-1) = 63/65$; the intake's exact coefficients give -0.9687 at $q_t = 40$ ($j_{q_t} = 0$) and $y_0(L) = 1/2$, against $-63/65 = -0.96923\dots$
- $r = 5$: K_q runs through all four nonzero residues, so J has four of the five residues, of both parities; there is no limit at any dyadic $\zeta \neq 1$, for every r_0 .
- $r = 6$: $K_q \equiv 2, 4$ alternately, so $J = \{1, 3\}$ for $r_0 = 0$: the limit exists at $\zeta = -1$ but not at $\zeta = i$. For $r_0 = 3, 4$ the two residues differ by 4, and the limit also exists at $\pm i$, but not at roots of order 8.

Near the radii where two blocks compete the section can vanish. For $d = 0$, $r = 3$, $r_0 = 0$ and $\zeta = -1$ the values $B_0^{(3,0)}(-R_0 e^{-t})$ are real, and by (109) they are $\omega_{q_t}(t)(-64/65 + o(1))$ when $j_{q_t} = 0$ and $\omega_{q_t}(t)(16/65 + o(1))$ when $j_{q_t} = 1$; since j_q alternates, the section has zeros on $(-R_0, 0)$ accumulating at $-R_0$, and $|B_0^{(3,0)}|$ does not tend to infinity along the whole radius. So the restriction $\delta \leq y_0(L) \leq 1 - \delta$ in Theorem 21.5 cannot be dropped. In the intake's computation at $q_t = 40$ the ratio $B_0^{(3,0)}(-R_0 e^{-t})/\omega_{q_t}(t)$ moves from -0.984 at $y_0(L) = 1/2$ to $+1.78$ as $y_0(L) \downarrow 0$, changing sign near $y_0(L) \approx 0.06$.

22 Computation, checks, and examples

22.1 What the computations test

Each manuscript ships its own program, written independently of the other and of Part I's (`code/03-source-extremum-verify.py` and `code/02-source-edge-verify.py`; the report's README gives the delivery names and a safe rerun). Both use exact integers for finite enumeration and identities, and both are error-detection companions to the proofs: no numerical result is used to prove a natural boundary, an asymptotic remainder, or the local L^p theorem. In their packages, manuscript 05's checks are reproduced by `python3 verify.py` from the package directory (its PDF by `pdflatex` twice or its `build.sh`), and manuscript 01's by `python code/verify.py` and `python code/make_article_tables.py` (or its `Makefile`); neither program accesses the network. In this report these files carry the prefixes of the provenance note, and running them in place would write into the report's directories; the README gives a rerun on a scratch copy.

Manuscript 05's program compares the marked-source sieve with the independent source-deletion recurrence (56), and also performs a small exhaustive graph enumeration; these are three different routes to the same finite objects. The exhaustive method examines every graph whose arcs point from a larger to a smaller index. It rejects repeated neighborhoods, collapses the survivors to finite sets, and deduplicates the resulting transitive sets. Every acyclic isomorphism class has such an ordering, so this exhausts the small cases independently of either recurrence.

Manuscript 01's program independently generates finite transitive sets: it starts with the empty transitive set and, from a transitive set F , adjoins each subset $x \subseteq F$ with $x \notin F$, deduplicating the resulting transitive sets $F \cup \{x\}$. Every nonempty finite transitive set can be reduced by deleting a source, so this generation is exhaustive. Nested immutable sets represent equality directly; no numerical coding with a truncated rank is used. Its exact arithmetic uses (15), (14) and (16), together with the deletion recurrence, and checks the initial core counts, the source maximum, the covering inequalities, the finite-defect formula and the elementary bounds (55).

The small rows $u(n, s)$, $n \leq 7$, are those of Part I's table in Section 11.1 (manuscript 05 prints them again, through $n = 7$, with $q(n)$ and $s_{\max}(n)$; that table is not repeated). In particular the source-maximizer counts begin 1, 1, 2, 1, 20, 20, 10, 2, 7128, 8316, 7128, 4455, 1980, 594, 108, 9 (Part I's table in Section 11.2). The jump from $b_8^{(0)} = 2$ to $b_9^{(0)} = 7128$ reflects a change from a three-vertex to a four-vertex core. At the radius $1/4$, the first few entries in each sufficiently late block provide most of its analytic mass, as quantified by the profile $1/(1 - z)$.

22.2 Finite verification and numerical diagnostics of manuscript 05

The exact verification covers both source triangles through $n = 64$, the source support and row sums, the finite-defect formula for $0 \leq d \leq 3$ in this range, the covering inequalities, and the

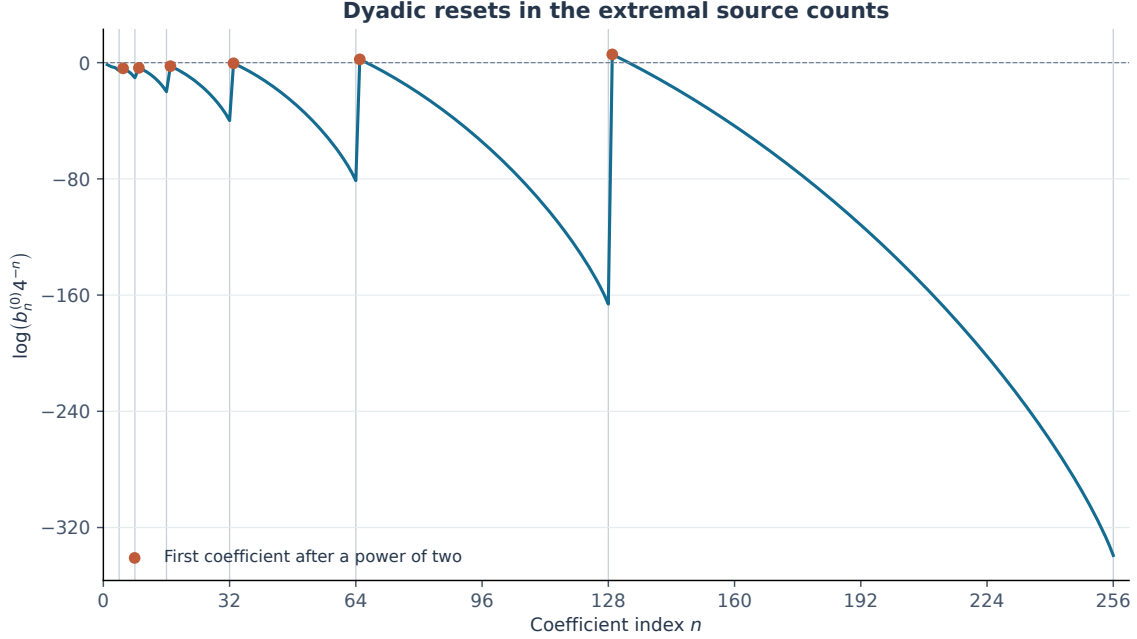


Figure 1: Normalized source-maximizer coefficients, computed from the exact formula $b_n^{(0)} = u_q \binom{2^q - q}{n - q}$ (manuscript 05). Positive coefficients occur throughout; the concentration just after dyadic block boundaries is the structure used by the proof.

core-count bounds. Exhaustive collapse enumeration through $n = 6$ independently checks the small rows. The machine-readable record `data/03-source-extremum-verification.json` gives the actual counts of checks and records any failure.

The recorded run compares 2,080 triangle cells and 242 admissible fixed-defect entries. Additional exact first-order product expansions check 1,040 coefficient instances of the block correction and 80 instances of the amplitude coefficient of Section 23. The independent six-vertex enumeration examines 32,768 ordered graphs, retains 3,240 extensional graphs, and deduplicates them to 1,802 isomorphism classes.

Numerical files examine the geometric block profile and the phase ratio in (48). The profile tables use the exact first-coefficient normalization

$$Q_q(z) = \sum_{j=0}^{K-1} \frac{b_{K+1+j}^{(d)}}{b_{K+1}^{(d)}} z^j = \frac{P_q(z)}{P_q(0)}.$$

The covering estimate gives $P_q(0) = 1 + O_d(m2^{-k_0})$, so Q_q has the same geometric limit and asymptotic correction orders as P_q . The numerical files distinguish exact integer count evaluations from binomial-proxy calculations for very large blocks. Floating-point diagnostics are not interval certificates. The covering inequality supplies the mathematical reason the proxy is appropriate; any reported finite numerical tolerance is only a diagnostic unless explicitly enclosed by a proved bound.

For a specified n and fixed d , (16) needs only $d + 1$ summands after $u_0, \dots, u_{q(n)+d}$ are tabulated; as Part I says in Section 11.3, this is an arithmetic-operation statement, not a claim of constant bit complexity. Exact integer arithmetic is appropriate for the sieve; logarithms and the positive proxy are more stable for asymptotic diagnostics near a boundary. The proofs themselves use only the coarse bounds (55), not a long table of large core counts.

Table 4: Numerical block profiles $Q_q(R_d)$, with $Q_q = P_q/P_q(0)$ normalized by the actual first coefficient $b_{K+1}^{(d)}$. The final row gives $1/(1 - a_d R_d)$. Retained coefficients are exact integers; evaluations are numerical.

q	$d = 0$	$d = 1$	$d = 2$
5	1.3796799967	1.6418450310	1.8627469784
8	1.3491920197	1.4971773509	1.5727604108
11	1.3365489305	1.4689177227	1.5315911859
14	1.3338949373	1.4638551844	1.5247758863
∞	1.3333333333	1.4628571429	1.5234819974

Table 5: Absolute errors in the numerical radial-ratio limit (48) at $r = 0.9999$. Large blocks use the covering proxy with analytic truncation and covering majorants. Floating-point roundoff is not interval-certified.

ζ	$d = 0$	$d = 1$	$d = 2$
i	2.445382×10^{-6}	7.869003×10^{-6}	1.223633×10^{-5}
$e^{\pi i/4}$	1.321876×10^{-6}	6.207385×10^{-6}	1.068750×10^{-5}

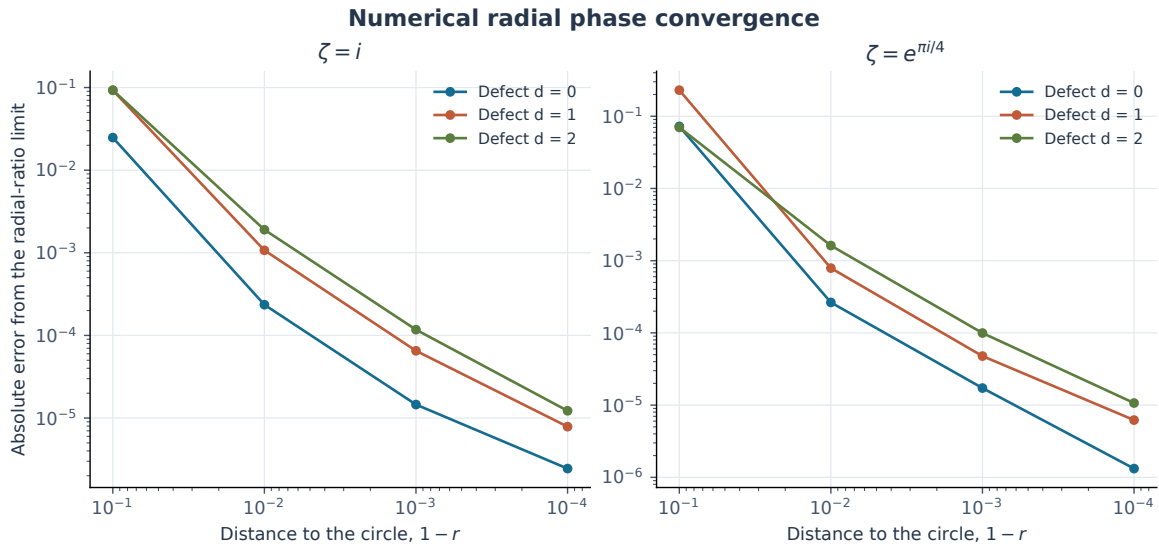


Figure 2: Numerical convergence to the complex phase limits for two dyadic roots (manuscript 05). The evaluations use exact coefficient prefixes and a covering-controlled proxy for large blocks. The plots are diagnostics, not proofs or interval-certified evaluations.

22.3 Analytic diagnostics of manuscript 01

Profile errors are evaluated at several roots of unity and compared with $1/(1 - aRw)$. This tests indexing, the normalization $u_m C_q$, and the constants R and aR ; the proof of Lemma 16.1 supplies the uniform closed-disk result. The radial calculations compare

$$\frac{B_d(R_d e^{-t} \zeta)}{B_d(R_d e^{-t})} \quad \text{with} \quad \frac{\zeta(1 - a_d R_d)}{1 - a_d R_d \zeta}$$

and record the logarithmic radial magnitude. For very large block starts, log-gamma evaluation and geometric truncation avoid constructing gigantic binomial coefficients. The recorded run uses 100 decimal working digits, exact coefficients through $n = 512$, and the first 128 offsets of each later block through $q = 60$. All $d + 1$ terms of the finite-defect formula are retained. Monotone consecutive ratios bound the omitted within-block terms, and the upper core bound in (55) bounds the infinite block tail. The largest evaluated relative truncation bound is 3.96×10^{-58} ; a 100-versus-140-digit endpoint comparison differs by less than 2.47×10^{-81} . These are floating-point evaluations of analytic bounds, not interval-arithmetic certificates enclosing roundoff. The data files retain the actual tested ranges, counts, and errors; a small sampled profile error is not being identified with the supremum over a complex disk. In the shipped table file below, $N = K$, $\rho_d = a_d R_d$ and $\varepsilon = t$.

22.4 Recorded exact results

n	u_n	Positive source row $u(n, 1), u(n, 2), \dots$
1	1	1
2	1	1
3	2	2
4	9	8, 1
5	88	68, 20
6	1,802	1,248, 534, 20
7	75,598	48,640, 25,008, 1,940, 10

The two recurrences agree on all 2,080 triangle cells through $n = 64$. The run also checks 242 positive finite-deficit entries against the triangle and 112 exact binomial spot checks of the adjacent-coefficient implementation.

22.5 A uniform profile diagnostic

At $q = 11$, $N = 1024$, the table gives the coefficient- ℓ^1 bound

$$\sum_{j=0}^{N-1} \left| \frac{b_{N+1+j}^{(d)}}{A_q} R_d^j - \rho_d^j \right| + \frac{\rho_d^N}{1 - \rho_d},$$

which bounds the closed-disk profile error. The displayed decimal values are high-precision evaluations of that bound, without an interval enclosure for floating-point roundoff.

d	Coefficient bound	Bound multiplied by N/m
0	0.003215632303	0.299346134
1	0.006060580187	0.517169509
2	0.008109188522	0.638754542
3	0.009672798955	0.707496152

22.6 A complex radial diagnostic

At $d = 0$ and $\varepsilon = 2^{-32}$, the absolute errors in (49) are approximately

ζ	-1	i	$e^{i\pi/4}$
absolute error	6.3523×10^{-12}	6.0306×10^{-12}	4.0592×10^{-12}

The recorded run contains 300 radial comparisons and 140 complex profile evaluations in total.

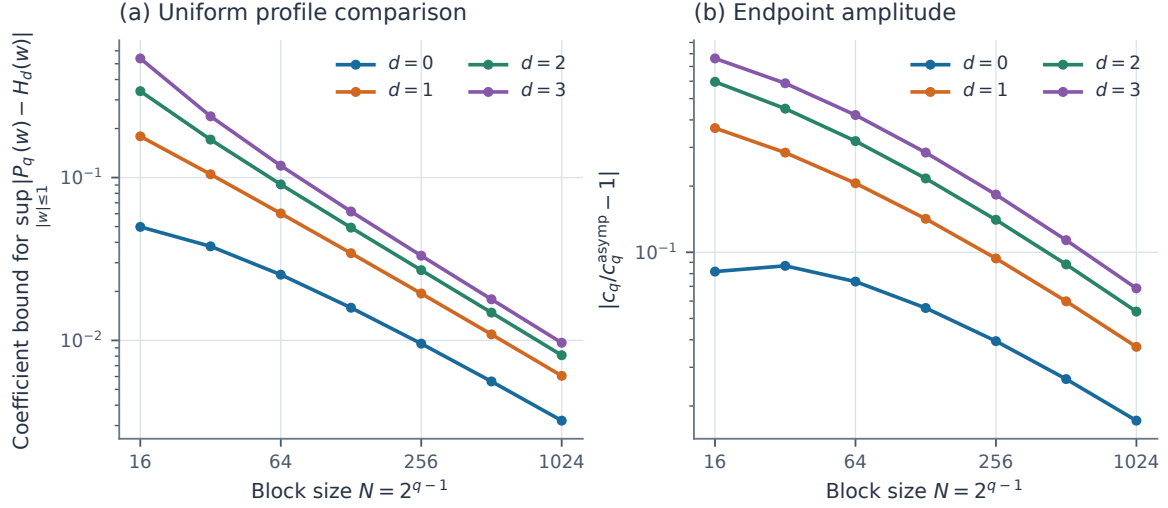


Figure 3: Diagnostics of the dyadic block profile (manuscript 01). Each finite computation tests the normalization and the geometric limiting profile $(1 - a_d R_d w)^{-1}$ (in the plot, $H_d(w) = (1 - \rho_d w)^{-1}$ with $\rho_d = a_d R_d$). The second panel compares the endpoint amplitude with the main term of (73). The closed-disk error estimate is proved in Lemma 16.1.

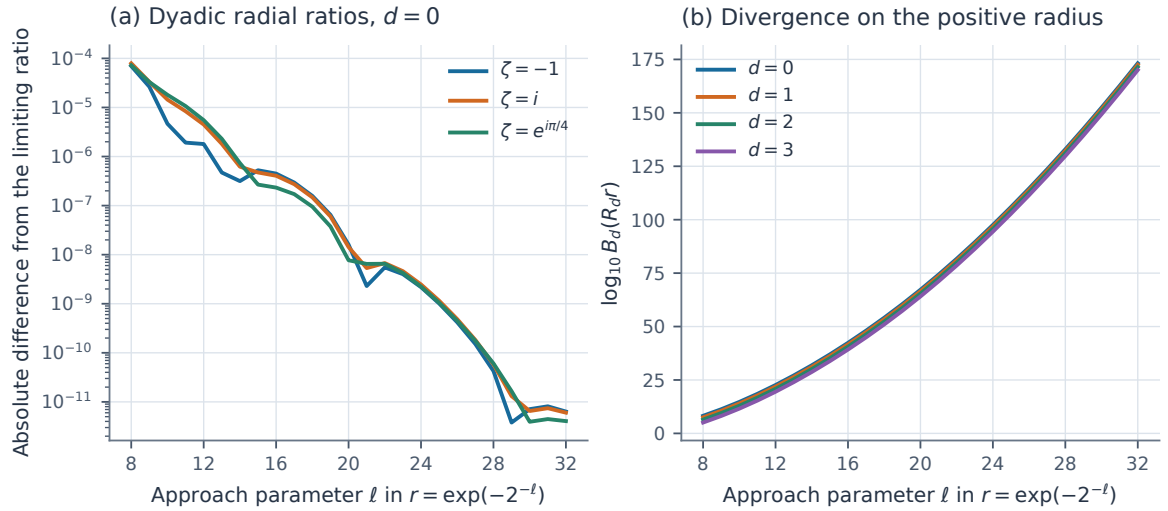


Figure 4: High-precision diagnostics of the radial limit and growth laws (manuscript 01). The exact limit constants and the two-term logarithmic expansion are proved in Section 18.

[write] Added 5 October 2026, batch 98: reruns and the write's checks. The writing step reran both programs on scratch copies in their delivered layouts (Python 3.12.13, mpmath 1.3.0,

Matplotlib 3.10.8, NumPy 2.3.5): manuscript 01's `verify.py` and `make_article_tables.py` with default parameters (about 77s here, against a recorded 15s), and manuscript 05's `verify.py--max-n64--brute-n6` (about 44s, against 10s). All fifteen CSV files were reproduced byte for byte, the three L^AT_EX table files up to Windows line endings; the JSON and text run records differ only in the Python patch version and the elapsed time. The three $u(n, k)$ tables shipped with this report (Part I's and both manuscripts') agree on all 1,759 positive cells with $n \leq 64$.

In a separate scratch computation (not shipped) the writing step checked the inputs of Section 21.2 and Proposition 24.1: the bounds (107) hold exactly for $m \leq 63$; with the exact amplitudes A_q for $d = 0, 1$ and $L \approx 15, 20, 25, 30$ chosen with $y_0(L) = 1/2$, the predicted block q_t is the largest term and $S_d(t)/\omega_{q_t}(t) = 1.023, 1.0035, 1.00065, 1.00009$; and for $d = 0$, step $r = 3$ and $L \approx 20.2$ ($q_t = 35$), the exact coefficients give $|B_0^{(3,0)}(\frac{1}{4}e^{-t}\zeta)|/\omega_{q_t}(t) = 0.2427$ at $\zeta = -1$ and 0.24999 at $\zeta = i$, against the values 0.2462 and 0.24997 of the bracket in (109) (at $L \approx 16.3$, $q_t = 29$, they are 0.2319 and 0.2504 ; at $L \approx 11.7$, $q_t = 22$, where j_{q_t} differs, 0.968 and 1.00003 against 0.985 and 0.9999). These are floating-point illustrations, not proofs.

Dated note (5 October 2026). The intake's check (item 13 of Section 24) reproduced these values and found two labels wrong. Under the definition (105), $L \approx 16.3$ and $L \approx 11.7$ give $q_t = 29$ and $q_t = 22$, not $q_t = 27$ and $q_t = 20$ as first printed; the printed values are those at $q_t = 29$ ($L = 16.272$) and $q_t = 22$ ($L = 11.746$), with $y_0(L) = 1/2$, and the labels are corrected above. At $q_t = 27$ and $q_t = 20$ the values would be $0.2236, 0.2510$ and $0.9587, 1.00023$. The values of j_{q_t} and of the brackets are unaffected.

23 First corrections inside a dyadic block

This section is manuscript 05's Appendix A. The natural-boundary proof needs only an $O(m/K)$ error. The following refinements make the mechanism more explicit and provide additional finite algebra for verification.

Proposition 23.1 (First block correction; source 05). *For every fixed $0 < T < 1/a$,*

$$P_q(z) = \frac{1}{1-az} + \frac{1}{K} \left\{ \frac{(m-1)az}{(1-az)^2} - \frac{cz}{(1-az)^3} \right\} + O_{d,T}(m^2/K^2), \quad |z| \leq T. \quad (110)$$

Proof. From (64), expansion of the logarithm gives

$$v_{q,j} = a^j \left[1 + \frac{1}{K} \left\{ j(m-1) - \frac{c}{2a} j(j+1) \right\} + \text{remainder} \right].$$

To justify summation uniformly, fix $\rho < 1$ with $(a + \epsilon)T < \rho$, where ϵ is as in the proof of Lemma 16.1. Split the sum at $J = \lfloor K^{1/4} \rfloor$. On $j \leq J$, Taylor's formula for each logarithm and then for the exponential bounds the weighted remainder by

$$\frac{C_{d,T}}{K^2} (m^2 j^2 + j^4 + m^2 j + j^3) \rho^j.$$

Here $m = O_d(\log K)$ ensures the expansions are uniform. The sum of these errors is $O_{d,T}(m^2/K^2)$. The tails $j > J$ of the exact product, the leading geometric series, and the polynomial correction are exponentially small in J , using the same ratio bound. The covering error is smaller still. Finally

$$\sum_{j \geq 0} j w^j = \frac{w}{(1-w)^2}, \quad \sum_{j \geq 0} j(j+1) w^j = \frac{2w}{(1-w)^3},$$

with $w = az$, give (110). □

(In this proof ρ is a local constant and w a local variable.)

Proposition 23.2 (First amplitude correction; source 05). *With the normalization (58),*

$$A_q = u_m c^{-m} R \sqrt{\frac{ca}{2\pi K}} \left[1 + \frac{E_{c,m}}{K} + O_d(m^4/K^2) \right], \quad (111)$$

where

$$E_{c,m} = (m-1) - \frac{a}{2c} m(m-1) + \frac{1/c - 1 - 1/a}{12}. \quad (112)$$

Proof. The product quotient in (70) has logarithm

$$\frac{1}{K} \left((m-1) - \frac{a}{2c} m(m-1) \right) + O_d(m^3/K^2).$$

Exponentiation changes the relative remainder to $O_d(m^4/K^2)$. Stirling's expansion gives

$$\left(\frac{cK}{K} \right) R^K = \sqrt{\frac{c}{2\pi a K}} \left[1 + \frac{1/c - 1 - 1/a}{12K} + O_d(K^{-2}) \right].$$

Multiply these formulas and the prefactor $u_m R a c^{-m}$. □

Neither proposition alone provides a further universal term in (48). Summing a first correction introduces weighted averages of m/K and $1/K$ over competing blocks. Lemma 17.2 controls their order, but a limiting coefficient would require additional information about those averages.

Table 6: First-order correction diagnostics at $z = R_d$. ε_P is the absolute remainder after the displayed first correction to the proxy-normalized profile P_q ; ε_A is the absolute remainder after $1 + E/K$ in the normalized proxy amplitude. The scalings illustrate the predicted orders; they are not proofs or interval certificates.

d	q	$(K^2/m^2)\varepsilon_P$	$(K^2/m^4)\varepsilon_A$
0	8	0.2079719	0.0005899
0	11	0.2836076	0.0041761
0	14	0.3371871	0.0080577
1	8	0.4691612	0.0178350
1	11	0.5593360	0.0272120
1	14	0.6297887	0.0340043
2	8	0.6637698	0.0348041
2	11	0.7404070	0.0462915
2	14	0.8121156	0.0541686

24 Further questions and research

The natural-boundary question of Part I is settled by the preceding proofs. The following questions merge the two manuscripts' lists (manuscript 05's Section 10, ten items, and manuscript 01's Section 11, nine items) with each other, with the related questions of Part I's Section 12.2, and with the claims of either manuscript that are argued but not proved. Each is credited; the sources' status is relative to their inspected repository and sources, not a claim that every formulation is historically open.

1. **The term of order L and the discrete saddle** (05, Q1; 01, Q1 and Section 7.3). Determine the next term after $\mathcal{T}(L)$ in (50), including its dependence on the position of the continuous maximizer relative to the integers. A sharper asymptotic for u_m can refine the block amplitudes (manuscript 01: combine a fully verified precise asymptotic for u_m with (73)), but the maximizing block index is an integer. Both manuscripts assert that rounding by a bounded amount changes the logarithm by order L , so that a smooth saddle alone cannot give a constant coefficient there, and ask for the correct phase variable and any periodic or switching behavior, and for an expression uniform through the crossover of two neighbouring dominant blocks. They argue this only heuristically (Remark 18.4). Proposition 24.1 below [write] proves it under a hypothesis on u_m and bounds the coefficient unconditionally. Missing: a proof of the hypothesis (114) (or of the oscillation without it), and the $o(L)$ terms.
2. **Pointwise behavior at nondyadic angles** (05, Q2; 01, Q2). The theorems prove a natural boundary everywhere and radial blow-up at dyadic angles. They do not say that every irrational angle has the same radial limit or magnitude. At a root of unity of odd order the block phases cycle; at an irrational angle they may behave much more irregularly. Characterize limsup and liminf radial growth there and at typical angles, retaining possible cancellation between blocks; determine whether radial magnitudes are unbounded and whether normalized limits exist on suitable subsequences.
3. **Angular transition scales and limiting profiles** (05, Q3; 01, Q3). Manuscript 05's proof uses a grid with spacing of order t/L and shows that large values occupy positive measure in every fixed arc; manuscript 01 says the dominant block index (87) "suggests" a phase scale of order t/L around a dyadic point. Neither proves a profile on that scale. Study the function on windows $\theta = \theta_0 + xt/L$, $e^{i\theta_0}$ dyadic, and on windows tied to individual block frequencies: is there an explicit limiting transition profile, or a family of subsequential profiles, including where two neighbouring blocks carry comparable mass? Lemma 21.4 identifies when one block dominates.
4. **Exact local L^p constants** (05, Q4). Theorem 14.4 gives comparison constants, not a limit for the ratio of the integral to $B_d(R_d e^{-t})^p$. Determine whether that ratio converges or has a phase-dependent set of limit points. Parseval's identity makes $p = 2$ a useful first target, but the competition between neighboring dyadic blocks must still be controlled.
5. **Arithmetic sections** (05, Q5; 01, Q4). For fixed integers $r \geq 2$ and $0 \leq r_0 < r$, determine the analytic continuation of $\sum_n b_{rn+r_0}^{(d)} z^n$. Manuscript 05 notes that a natural boundary of the whole series does not automatically pass to a section: root-of-unity filters can introduce cancellation, and the block starting residues depend on q . Manuscript 01 proves the natural boundary for every step 2^s (Theorem 21.1), and Theorem 21.5 [write] proves it for every step. For a step with an odd factor, Corollary 21.7 decides whether the radial ratio at a dyadic root has a limit. Still open for such a step: the behaviour near radii where two blocks compete, beyond the sign changes noted after that corollary. Manuscript 01's question which phase alignments force dense singularities is answered at the level of natural boundaries.

Dated note (5 October 2026). Until this date the item read: "Still open for a step with an odd factor: whether the radial ratio at a dyadic root has a limit, or only the subsequential values that (109) suggests, and the behaviour near radii where two blocks compete." The intake's check (item 13) found that Theorem 21.5 already answers the first part: the ratio has a limit exactly when ζ^{jq} is eventually constant, and otherwise its distinct subsequential limits are the values $\zeta^{1+j}(1 - (a_d R_d)^r)/(1 - (a_d R_d \zeta)^r)$ (Corollary 21.7). For $r = 3$, $r_0 = 0$ there is no limit at any dyadic $\zeta \neq 1$ (for $d = 0$ the ratio at $\zeta = -1$ jumps between $-63/65$ and $63/65$); for $r = 6$, $r_0 = 0$ the limit exists at -1 but not at i . That part is therefore no longer listed as open.

6. **Growing defects** (05, Q6; 01, Q5; Part I, Question 3). Find useful uniform ranges for the counting and block estimates when d grows with q or n . The covering estimate remains available, and the simple condition $m = q + d = o(K)$ still favors the geometric profile, but the radius R_d moves toward zero and the block geometry fixes $c = 2^{d+1}$; the radius and even the right normalization may change, so this needs a fresh formulation rather than a substitution in an O_d bound. Theorem 20.3 rules out an ordinary holomorphic germ with positive defect marking; it does not rule out appropriate formal, rescaled, or summability-based descriptions.
7. **Complex defect fugacity** (05, Q7). Analyze $\sum_d b_n^{(d)} v^d$ for negative or nonreal v . The positive-fugacity proof cannot decide the effect of cancellation. Exact marked-source identities may reveal special values with substantially different growth.
8. **Nonlinear differential algebraicity** (05, Q8; 01, Q9; Part I, Remark 8.3). Does some B_d satisfy a nonlinear polynomial differential equation over $\mathbb{C}(z)$; are these series differentially algebraic or differentially transcendental? Natural boundaries prohibit linear differential equations here, but are not by themselves an obstruction to nonlinear differential algebraicity. No conclusion of Part II settles it.
9. **Other neighborhood capacities** (05, Q9; Part I, Question 8). For bounded outdegree Δ , replace the Boolean capacity 2^m by $\sum_{j \leq \Delta} \binom{m}{j}$. Determine which source boundaries are sharp and whether the resulting block spacing still yields complex noncontinuation. Fixed Δ and growing Δ may behave differently because their capacity scales differ.
10. **Weighted source and arc statistics** (01, Q6; Part I, Questions 5 and 6). Mark the number of arcs or the source outdegree profile while counting graphs at a fixed source defect. Derive a multivariate analogue of the Boolean-core profile, and study the corresponding singularity sets as the marking parameter varies. The deleted core neighborhoods are a small but structured correction to a Boolean subset model.
11. **Certified computation** (05, Q10; 01, Q7). Turn the explicit ratio and covering inequalities into an interval-arithmetic evaluator near a specified dyadic boundary point, combining exact core counts with explicit tail inequalities to enclose complex values and phase-ratio errors rigorously, and selecting the core cutoff, block cutoff and within-block truncation from a requested tolerance. The present code detects mistakes; it is not such a certificate. This is not needed for the analytic proofs.
12. **Formal verification** (05, Q10; 01, Q8; Part I, Question 10 and Section 12.1). Turn the exact core recurrence, marked-source bijection, and covering estimate into reusable checked lemmas. A separate analytic development could formalize the positive dyadic-sum estimate and the dense-grid argument; manuscript 01 names the general Lemma 18.1, followed by the coefficientwise product bound (66), as a compact first target, with the finite set and source-counting lemmas a separate exact component.
13. **Independent review of Part II** (05, README and source notes; the writing step). Manuscript 05's delivery README, its source notes and its ledger say that "independent internal reviews" or "separate derivations" checked the core counting, the block estimates, the radial transfer and the moving-grid argument. That is a statement about the manuscript's preparation which the intake could not verify; it is recorded as the source's statement, not as evidence. The intake read the main proofs of both manuscripts and found no gap, and the two manuscripts prove the shared theorems independently, but neither is a review. Missing: an external mathematical review of Part II, including the writing step's Lemma 21.4, Theorem 21.5 and Proposition 24.1.
Dated note (5 October 2026). An independent adversarial check of the results added at the write, made by the intake after the write, found Lemma 21.4, Theorem 21.5 and Proposition 24.1 valid, with Remark 24.2 reproduced. It reread every step of the three proofs and tested them against independent exact computations. The bounds (107) and the

marked-source identity behind them hold exactly for $m \leq 63$, where all 2,145 cells of the shipped triangle agree with the sieve. The ratios S_d/ω_{q_t} of the note at the end of Section 22 were reproduced from the exact amplitudes for $d = 0, 1$. The asymptotic (109) was tested with exact coefficients $b_n^{(d)}$ (n up to 2^{52}) for $d = 0$, steps $r = 2, 3, 5, 6$, every residue r_0 and $\zeta \in \{1, -1, i, e^{i\pi/4}, e^{3i\pi/8}\}$ at $y_0(L) = 1/2$: the largest absolute error inside the bracket falls from 0.11 at $q_t = 20$ to $2.4 \cdot 10^{-4}$ at $q_t = 46$, and for $d = 1$ ($r = 3, 5, 6$) it is $9 \cdot 10^{-5}$ at $L \approx 30.2$. For large r the relative error can stay large at moderate L when $j_{q_t} = r - 1$ (0.90 for $r = 6$, $\zeta = e^{i\pi/4}$, $q_t = 35$, where the bracket is about 4^{-5}); the theorem's $o(1)$ is absolute, so this is allowed. An adversarial probe at the crossover radii ($d = 0$, $r = 3$, $r_0 = 0$, $\zeta = -1$) found the section changing sign near $y_0(L) \approx 0.06$, which shows that the theorem's restriction on $y_0(L)$ is necessary (the note after Corollary 21.7 derives this from the theorem). For Proposition 24.1 the check confirmed the envelope (116) against the exact amplitudes for $q \leq 64 - d$, $d = 0, 1, 2$, the constants of Remark 24.2 ($\log(u_{m+1}/u_m) - hm$ lies in $[-0.3974857226, -0.3974857210]$ for $30 \leq m \leq 63$, and $\alpha/h - 3/2 \approx -2.5735$), and the predicted term $(\alpha/h - 3/2 + \varphi(y_0(L)))L$ in exact sums for $d = 0, 1, 2$ and $7 \leq L \leq 36$. It found one understatement, corrected by Corollary 21.7 and the dated note in item 5, and two wrong labels q_t in the note at the end of Section 22, corrected there. The check was a careful reading with computations, not a formal verification or an external review, so this item stays open.

Proposition 24.1 (The order- L term; write). *Let $\varphi(y) = 1 + y - 2^y$ and let $y_0(L)$ be as in (105). Fix $d \geq 0$. As $t \downarrow 0$, with $L = \log(1/t)$:*

(i) *unconditionally,*

$$(-3 + \varphi(y_0(L)))L - o(L) \leq \log B_d(R_d e^{-t}) - \mathcal{T}(L) \leq (-2 + \varphi(y_0(L)))L + o(L); \quad (113)$$

(ii) *if, for a constant α ,*

$$\log u_m = \frac{h}{2}m^2 + \alpha m + o(m) \quad (m \rightarrow \infty), \quad (114)$$

then

$$\log B_d(R_d e^{-t}) = \mathcal{T}(L) + \left(\frac{\alpha}{h} - \frac{3}{2} + \varphi(y_0(L))\right)L + o(L). \quad (115)$$

On $[0, 1]$, $\varphi \geq 0$ vanishes only at the endpoints and attains its maximum $\varphi^* = 1 - (1 + \log \log 2)/\log 2 = 0.08607 \dots$ at $y^* = -\log_2 \log 2$, and $y_0(L)$ takes every value in $[0, 1]$ along sequences $L \rightarrow \infty$. Hence, under (114), $(\log B_d(R_d e^{-t}) - \mathcal{T}(L))/L$ has $\liminf$ equal to $\alpha/h - 3/2$ and $\limsup$ equal to $\alpha/h - 3/2 + \varphi^*$: the remainder of (50) has no fixed coefficient of L , but oscillates with the fractional part of $(L + \log L)/\log 2$. The same holds, with the same constants and independently of d , for every fixed $d \geq 0$.

Proof. By (82), $\log B_d(R_d e^{-t}) = \log S_d(t) + \log g(R) + o(1)$, so it suffices to treat $S_d(t) = \sum_q \omega_q(t)$. By (73), $\log A_q = \log u_{q+d} - (q+d)(d+1)h - \frac{1}{2}(q-1)h + O_d(1)$. Inserting the bounds (55), $\frac{h}{2}(m-1)(m-2) \leq \log u_m \leq \frac{h}{2}m(m-1)$ with $m = q+d$, gives

$$\frac{h}{2}q^2 - 3hq - C_d \leq \log A_q \leq \frac{h}{2}q^2 - 2hq + C_d, \quad (116)$$

and (114) gives $\log A_q = \frac{h}{2}q^2 + (\alpha - \frac{3}{2}h)q + o(q)$. Since S_d is increasing in each A_q , both parts follow from the claim: *if $\log A_q = \frac{h}{2}q^2 + \beta q + e_q$ with $e_q = o(q)$ and $|e_q| \leq C'q$, then $\log S(t) = \mathcal{T}(L) + (\beta/h)L + L\varphi(y_0(L)) + o(L)$.*

To prove the claim write $X = (L + \log L)/h$ and $q = 1 + X + y$, so that y runs through $y_0(L) + \mathbb{Z}$, and $t(K_q + 1) = L2^y + t$. Expanding $\frac{h}{2}q^2$ and βq gives

$$\log \omega_q(t) = \mathcal{T}(L) + \frac{\beta}{h}L + L\varphi(y) + \frac{h}{2}(1+y)^2 + (1+y)\log L + \beta(1+y) + \frac{(\log L)^2}{2h} + \frac{\beta}{h}\log L - t + e_q. \quad (117)$$

For the lower bound take the single term $y = y_0(L) \in [0, 1]$; all terms after $L\varphi(y)$ are then $O((\log L)^2) + o(L)$. For the upper bound, φ is concave with $\varphi(0) = \varphi(1) = 0$, so $\varphi \leq 0$ off $[0, 1]$, and $y_0(L)$ is the only point of $y_0(L) + \mathbb{Z}$ in $[0, 1]$; hence $\varphi(y) \leq \varphi(y_0(L))$ for every admissible y . For $|y| \leq Y$, a fixed bound, there are at most $2Y + 2$ terms, each at most $\mathcal{T}(L) + (\beta/h)L + L\varphi(y_0(L)) + o(L)$. For $y > Y \geq 3$, $\varphi(y) \leq -2^{y-1}$, while the other terms of (117) are $O((1+y)^2 + (1+y)\log L + (\log L)^2) + C'(X+1+y)$; for Y with $2^{Y-2} > C'/h + 2$ and large L these terms sum to at most $\exp(\mathcal{T}(L) + (\beta/h)L - 2L)$. For $y < -Y$, $1+y < 0$ and $q \geq 1$ gives $|1+y| \leq X$; then $L\varphi(y) + (1+y)\log L + \frac{h}{2}(1+y)^2 \leq (1+y)(L + \log L + \frac{h}{2}(1+y)) \leq -\frac{1}{2}|1+y|L$, so the term is at most $\mathcal{T}(L) + (\beta/h)L - |1+y|(L/2 - |\beta| - C') + C'X + O((\log L)^2)$; there are at most $X + 1 = O(L)$ such terms, and for Y with $(Y-1)/3 > C'/h + 2$ their sum is again at most $\exp(\mathcal{T}(L) + (\beta/h)L - 2L)$ for large L . This proves the claim. Part (i) follows with $\beta = -3h$ and $\beta = -2h$ (e_q bounded), part (ii) with $\beta = \alpha - \frac{3}{2}h$. Finally $\varphi'(y) = 1 - h2^y$ vanishes at $y^* = -\log_2 h \in (0, 1)$, where $\varphi(y^*) = 1 - \log_2 h - 1/h = \varphi^*$, and $y_0(L) = \lceil X \rceil - X$ decreases continuously from near 1 to 0 on each interval between consecutive integer values of X . $\square$

Remark 24.2 (Numerical evidence for (114); write). The bounds (55) only give $\alpha \in [-3h/2, -h/2]$ if α exists. From the exact values of u_m shipped with this report ($m \leq 64$), $\log(u_{m+1}/u_m) - hm$ equals $-0.3974857\dots$ to eight digits for $30 \leq m \leq 63$, which suggests (114) with $\alpha = -0.3974857\dots - h/2 \approx -0.7441$, hence a coefficient of L oscillating between about -2.574 and -2.487 for every d . This is a numerical observation, not a proof; Wagner's asymptotic enumeration [8, 10], which the intake did not read, may settle it. A synthetic check with $A_q = \exp(\frac{h}{2}q^2 + \beta q)$ at L up to $2 \cdot 10^5$ reproduces (115), with the difference between the two sides of order $(\log L)^2/L$.

25 Conclusions, source audits and proof-status ledger

25.1 The mechanism (source 01)

The mechanism behind the answer has three parts. The source-covering estimate reduces exact graph counts to a binomial expression with a uniformly tiny relative error. The normalized binomial expression has a geometric profile across each late dyadic block. At dyadic roots of unity, the block starting exponents eventually have a common phase. The block amplitudes diverge, so the common profile forces nonzero radial growth at a dense set of boundary points. The elementary chain construction determines the quadratic logarithmic size of those amplitudes, and a discrete exponential-sum estimate then gives the first two radial growth terms. The arc-mean and subsequence results follow from the additional arguments stated above.

For a reader interested only in the natural boundary, manuscript 01 names the necessary chain: the finite collapse (Lemma 2.1) and the intrinsic core identity (8); the covering estimate (54), giving the exact-to-binomial comparison; Lemma 15.2 and Proposition 16.3, giving $A_q \rightarrow \infty$; the uniform profile (67) on the closed disk; and Lemma 18.1, giving dense radial singularities. The exact sieve is used for reproducibility and contextual counting. The full entropy theorem of Part I, a gap theorem, and a precise unrestricted asymptotic formula for u_m are not premises of this proof.

25.2 The repository snapshot and the literature checked

Both manuscripts inspected ProveIt at commit

d18416ec7e187a0948248cb3077e37d7b089a8f9,

reading this report's `article.tex` and README (then the text of Part I) under

```
SetTheory/Cardinals/docs/reports/
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a182220-source-boundary/,
```

manuscript 05 also the collection's report inventory and manuscript 01 a search of the collection's sources for the sequence identifiers and the natural-boundary question. Both identify the target as Section 12.2, Question 1 (Section 12.2 here), and manuscript 05 notes that the proof-status ledger records the natural boundary as not proved. Neither modified the repository; the inspection records the comparison made, not an exhaustive search of the literature.

The OEIS entries A182220, A182162, A001192 (and, for manuscript 01, A182161) were read on 5 October 2026 (UTC). A182220 still labels the displayed logarithmic formulas conjectural in that retrieved version. The source maximum is not presented as a new discovery: the bound predates this report, and Part I already proves the matching construction. The new target is the complex continuation of the fixed-defect enumerations. The numerical tables of both packages are computed by their own code, not copied from OEIS.

Tomescu's publicly available thesis was read at its counting chapter, including the source bound and the deletion recurrence. The publisher abstract of Policriti–Tomescu was inspected, as were the OEIS enumeration entries. Wagner's unrestricted asymptotic work is cited as context; none of its sharper results is an input to a proof here (manuscript 01 read the abstract and bibliographic record of the journal version, whose full text it could not access). Breuer–Simon's author text was read for the definition of a strong natural boundary and the scope of its general methods. The NIST DLMF supplies the Stirling expansion with its positive real remainder [12]. A bounded search for extensional-digraph natural boundaries did not locate a prior treatment of the present fixed-defect question; this is a statement about the search, not a theorem about priority. The verified claim of progress is that the exact question posed in Part I is answered by the proofs above.

25.3 Status ledger of Part II

Result	Evidence and scope
Finite collapse, rigidity, source support, marked-source sieve	Part I's, classical or repository background; re-proved by both manuscripts.
Sieve, recurrence, fixed-defect formula	Exact bijective and algebraic proofs; finite checks use independent methods.
Deletion recurrence	Bijective proof (Proposition 15.4), new to the report; the recurrence is Tomescu's.
Core envelope and covering bound	Elementary inequalities proved for the full stated ranges; the chain lower bound, with no priority claim.
Uniform block profile	Quantitative product and tail estimates; uniform on a disk larger than $ z = R_d$ (05); error $O_d(q2^{-q})$ on the closed disk (01).
Dyadic-sum growth and concentration	Explicit discrete estimates; no sharp core asymptotic assumed.
Natural boundary for every fixed defect	Proved by both manuscripts; answers Part I's Question 1.
Universal dyadic radial phase, with $O(t)$ rate	Proved for every fixed d and every fixed dyadic root (05; the limit also by 01's transfer lemma).
Two-term logarithmic radial growth	Proved with $O(L)$ remainder (both).
Local L^p comparison on every arc	Moving-grid and positivity proof, valid for every fixed $p > 0$ and open arc (05); the log-level law also by a single-point proof (01).
Power-of-two arithmetic sections	Proved (01, Theorem 21.1).
Positive-fugacity zero radius	Proved for real $v > 0$; no arbitrary complex- v claim.
First block and amplitude corrections	Taylor expansions with uniform remainders (Section 23).
Single dominant block; sections of every step; the order- L term	The writing step's Lemma 21.4, Theorem 21.5 and Proposition 24.1: proofs included; found valid by the intake's adversarial check after the write (5 October 2026; Section 24, item 13), not externally reviewed; part (ii) of the proposition is conditional on (114).
Radial ratios of sections at dyadic roots	Corollary 21.7 (the intake's, 5 October 2026): a limit exactly when ζ^{jq} is eventually constant, derived from Theorem 21.5.
Exact finite enumeration and recurrence checks	Reproducible computational checks, with recorded output; rerun at intake.
Numerical diagnostics, profile and radial plots	Reproducible error-detection experiments; floating-point values are not certified enclosures.
Independent mathematical checking	Manuscript 05 reports separate derivations of the core bound, profiles, radial estimates and local L^p argument; this is its own statement, not external peer review (Section 24, item 13).
Lean or Rocq verification	Not performed; no such certification is claimed.

Result	Evidence and scope
Historical priority of the new analytic refinements	Not established by the limited literature searches.

Appendices to Part I

A Suggested concise OEIS-facing statements

A182220. For every $n \geq 1$, $a(n) = n - \lceil \log_2 n \rceil$. Proof: if k vertices are sources, all n distinct out-neighborhoods are subsets of the $n - k$ nonsources, so $n \leq 2^{n-k}$. Conversely, take an extensional acyclic core on $m = \lceil \log_2 n \rceil$ vertices and choose $n - m$ unused subsets as new source neighborhoods. Since $n > 2^{m-1}$ for $m \geq 1$, any resulting family covers the core, so it has exactly $n - m$ sources. The case $n = 1$ is immediate. The upper bound already appears in Tomescu's 2011 thesis, printed page 29.

A182162. Let $T(n, k)$ denote the labeled count. For $q = \lceil \log_2 n \rceil$,

$$T(n, n - q) = n! \text{A001192}(q) \binom{2^q - q}{n - q}.$$

More generally, for $d \geq 0$ and $k = n - q - d \geq 1$, formula (16), multiplied by $n!$, gives an exact $(d + 1)$ -term expression. These are specializations of the marked-source sieve, with the extremal case also admitting the direct Boolean-core bijection.

Boundary sequence for a possible separate entry. The unlabeled extremizer sequence begins

$$1, 1, 2, 1, 20, 20, 10, 2, 7128, 8316, 7128, 4455, 1980, 594, 108, 9, \dots$$

It satisfies $b_{2^m}^{(0)} = \text{A001192}(m)$, $\liminf (b_n^{(0)})^{1/n} = 1$, and $\limsup (b_n^{(0)})^{1/n} = 4$, and it is not P-recursive. No assertion is made that this sequence currently lacks an OEIS entry; a sequence-value and literature search should precede any new-entry proposal.

[write] **Added 3 October 2026, batch 85.** This appendix is draft text. Nothing in it was submitted to OEIS, by the package's author (question 10 of Section 12.2) or by the intake, and this report submits nothing. Any proposal must keep Tomescu's credit for the upper bound and is a human editor's decision after independent review. The intake did not search OEIS for the boundary sequence displayed above.

B Proof-status ledger

Statement	Status and qualification
Maximum-source upper bound	Known in Tomescu (2011); independently proved here.
Maximum formula and full support	Proved constructively here; supplies a proof of the displayed OEIS conjecture. No historical-first claim.
Finite collapse and rigidity	Classical; proofs included to fix semantics and labeling factors.
General source sieve	Existing enumeration; proof included and attribution retained.
Exact extremizer count	Proved by a classification and also by specialization of the sieve; no priority certification.
Finite-defect formula	Exact specialization with a proved $d + 1$ -term truncation.
Uniform covering bound	Proved by a corewise probability estimate.
Entropy law and all limit rates	Proved uniformly for each fixed defect using elementary bounds on u_m .
Non-P-recursiveness	Proved for every fixed defect, including labeled counts.
All-orders expansion	Proved away from $p = 1$, with the core amplitude retained exactly.
Boundary binomial arc law	Proved in total variation with an explicit finite bound.
Numerical verification	Exact recurrence checks through 64 and exhaustive class enumeration through 7; not a substitute for the proofs.
Lean or Rocq verification	Not performed; no such certification claimed.
Natural boundary for B_d	Not proved; proposed as a further question.
Historical priority of refinements	Not established by the available search.

[write] **Added 5 October 2026, batch 98.** This ledger describes Part I. Its row “Natural boundary for B_d : not proved” is superseded: Part II proves the natural boundary for every fixed d (Theorem 14.2), in two independent unrefereed manuscripts and without formal verification. Part II has its own ledger (Section 25).

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- [10] Stephan Wagner, *Asymptotic enumeration of extensional acyclic digraphs*, Algorithmica **66**(4) (2013), 829–847, <https://doi.org/10.1007/s00453-012-9725-4>. Expanded version of [8]; cited by both Part II manuscripts as context only (manuscript 01 inspected the publisher’s abstract and bibliographic record).
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<https://doi.org/10.1016/j.aim.2010.12.019>; author text
<https://arxiv.org/abs/1002.0823>. Used in Part II for terminology (strong natural boundary) only.
- [12] NIST Digital Library of Mathematical Functions, *Section 5.11: Asymptotic Expansions of the Gamma Function*, especially formulas 5.11.1–5.11.3 and the real positive remainder discussion in Section 5.11(ii), <https://dlmf.nist.gov/5.11>.